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Technical Specification

O-RAN Working Group 7 Whitebox Hardware
Hardware Reference Design Specification for Outdoor Macrocell with Split Architecture Option 7.2

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Foreword

This Technical Specification (TS) has been produced by O-RAN Alliance.

Modal verbs terminology

In the present document "**shall**", "**shall not**", "**should**", "**should not**", "**may**", "**need not**", "**will**", "**will not**", "**can**" and "**cannot**" are to be interpreted as described in clause 3.2 of the O-RAN Drafting Rules (Verbal forms for the expression of provisions).

"**must**" and "**must not**" are **NOT** allowed in O-RAN deliverables except when used in direct citation.

Chapter 1 Introductory Material

1.1 Scope

This Technical Specification has been produced by the O-RAN.org.

The contents of the present document are subject to continuing work within O-RAN WG7 and may change following formal O-RAN approval. Should the O-RAN.org modify the contents of the present document, it will be re-released by O-RAN Alliance with an identifying change of release date and an increase in version number as follows:

Release x.y.z

where:

x the first digit is incremented for all changes of substance, i.e., technical enhancements, corrections, updates, etc. (the initial approved document will have x=01).

y the second digit is incremented when editorial only changes have been incorporated in the document.

z the third digit included only in working versions of the document indicating incremental changes during the editing process. This variable is for internal WG7 use only.

The present document specifies system requirements and high-level architecture for the FR1 Outdoor Macrocell deployment scenario as specified in the Deployment Scenarios and Base Station Classes document [1].

In the main body of this specification (in any “chapter”) the information contained therein is informative, unless explicitly described as normative. Information contained in an “Annex” to this specification is always informative unless otherwise marked as normative.

1.2 References

The following documents contain provisions which, through reference in this text, constitute provisions of the present document.

[1] ORAN-WG7.DSC.0-V03.00 Technical Specification, ‘Deployment Scenarios and Base Station Classes for White Box Hardware’. https://www.o-ran.org/specifications.

[2] 3GPP TR 21.905: "Vocabulary for 3GPP Specifications".

[3] 3GPP TS 38.104: "NR; Base Station (BS) radio transmission and reception".

[4] ORAN-WG4.CUS.0-v01.00 Technical Specification, ‘O-RAN Fronthaul Working Group Control, User and Synchronization Plane Specification’. https://www.o-ran.org/specifications

[5] 3GPP TS 38.113:"NR: Base Station (BS) Electromagnetic Compatibility (EMC)". http://www.3gpp.org/ftp//Specs/archive/38_series/38.113/38113-f80.zip

[6] O-RAN-WG7 OMAC Hardware Architecture Description - v02.00 Technical Specification 'Outdoor Macrocell Hardware Architecture and Requirements (FR1) Specification'

[7] https://www.rosenberger.com/0_documents/de/catalogs/ba_communication/catalog_coax/15_Chapter_43-10_41-95.pdf

[8] https://www.commscope.com/product-type/cable-assemblies/wireless-cable-assemblies/coaxial-cable-assemblies/itemmloc-f1xm-4/

[9] Zhong, J., Liu, D., Li, Z., & Sun, X. (2012, August). High thermal conductivity materials and their application on the electronic products. In 2012 IEEE Asia-Pacific Conference on Antennas and Propagation (pp. 173-175). IEEE.

[10] https://www.engineersedge.com/heat_transfer/convective_heat_transfer_coefficients__13378.htm

[11] https://www.thermoworks.com/emissivity-table/

[12] "Serial Interface for Data Converters", JESD204C.1, December 2021, www.jedec.org

[13] "JESD204x Frame Mapping Table Generator", https://www.analog.com/en/license/licensing-agreement/jesd204x-frame-mapping-table-generator.html

1.3 Definitions and Abbreviations

1.3.1 Definitions

For the purposes of the present document, the terms and definitions given in 3GPP TR 21.905 [1] and the following apply. A term defined in the present document takes precedence over the definition of the same term, if any, in 3GPP TR 21.905 [2]. For the base station classes of Pico, Micro and Macro, the definitions are given in 3GPP TR 38.104 [3].

Carrier Frequency: Center frequency of the cell.

F1 interface: The open interface between O-CU and O-DU defined by 3GPP TS 38.473.

Frequency Band: A designated frequency range for the operation of the base station and the UE radios. 5G NR frequency bands are divided into two different frequency ranges: Frequency Range 1 (FR1) that mainly includes sub-6GHz frequency bands, some of which are bands traditionally used by previous standards but has been extended to cover potential new spectrum offerings from 410MHz to 7125MHz; Frequency Range 2 (FR2) that includes frequency bands from 24.25 GHz to 52.6 GHz. Bands in this millimeter wave range have shorter range but higher available bandwidth than bands in the FR1.

Frequency Range: It refers to bandwidth defined by the frequency range within which the Base Station can be operated, defined by the band-pass filter of the BS; e.g., 3.4 – 3.8 GHz (400 MHz)

Fronthaul Gateway (FHGW): A fronthaul gateway is a physical entity that is located between a distributed unit and one or more radio units where it distributes, aggregates, and/or converts fronthaul protocols between the distributed unit and multiple radio units.

gNB: A RAN node providing NR user plane and control plane protocol terminations towards the UE, and connected via the NG interface to the 5GC.

Integrated architecture: In the integrated architecture, the O-RU and O-DU are implemented on one platform. Each O-RU and RF front end is associated with one O-DU. They are then aggregated to O-CU and connected by F1 interface.

Split architecture: The O-RU and O-DU are physically separated from one another in this architecture. A switch may aggregate multiple O-RUs to one O-DU. O-DU, switch and O-RUs are connected by the front haul interface as defined in WG4.

1.3.2 Abbreviations

For the purposes of the present document, the abbreviations given in [2] and the following apply. An abbreviation defined in the present document takes precedence over the definition of the same abbreviation, if any, as in [2].

7-2x	Fronthaul interface split option as defined by O-RAN WG4, also referred to as 7-2x
3GPP	Third Generation Partnership Project
5G	Fifth-Generation Mobile Communications
ADC	Analog to Digital Converter
AFE	Analog Front End
AFU	Antenna Filter Unit
ASIC	Application Specific Integrated Circuit
BBDEV	Baseband Device
BH	Backhaul
BMC	Baseboard Management Controller
BPSK	Binary Phase Shift Keying
BS	Base Station
CISPR	International Special Committee on Radio Interference
CFR	Crest Factor Reduction

CU	Centralized Unit as defined by 3GPP
COM	Cluster Communication
CPRI	Common Public Radio Interface
CPU	Central Processing Unit
CRC	Cyclic Redundancy Check
DAC	Digital to Analog Converter
DDC	Digital Down Conversion
DDR	Double Data Rate
DFE	Digital Front End
DIMM	Dual-Inline-Memory-Modules
DL	Downlink
DPD	Digital Pre-Distortion
DPDK	Data Plane Development Kit
DSP	Digital Signal Processor
DU	Distributed Unit as defined by 3GPP
DUC	Digital Up Conversion
ECC	Error Correcting Code
EMI	Electromagnetic Interference
eCPRI	evolved Common Public Radio Interface
EMC	Electro Magnetic Compatibility
EVM	Error Vector Magnitude
FCC	Federal Communications Commission
FEC	Forward Error Correction
FFT	Fast Fourier Transform
FH	Fronthaul
FHGW	Fronthaul Gateway
$FHGW_x$	Fronthaul gateway with no FH protocol translation, supporting an O-DU with split option x and an O-RU with split option x, with currently available options 6→6, 7-2x→7-2x and 8→8

$FHGW_{x \rightarrow y}$	Fronthaul Gateway that can translate fronthaul protocol from an O-DU with split option x to an O-RU with split option y, with currently available option 7-2$x_{\rightarrow}$8.
FHHL	Full Height Half Length
FPGA	Field Programmable Gate Array
GbE	Gigabit Ethernet
GNSS	Global Navigation Satellite System
GPP	General Purpose Processor
GPS	Global Positioning System
HARQ	Hybrid Automatic Repeat request
HHHL	Half Height Half Length
IP67	Ingress Protection 67
IEEE	Institute of Electrical and Electronics Engineers
IFFT	Inverse Fast Fourier Transform
IMD	Inter Modulation Distortion
I/O	Input/Output
JTAG	Joint Test Action Group
L1	Layer 1
LDPC	Low-Density Parity Codes
LDO	Low dropout
LRDIMM	Load-Reduced Dual In-line Memory Module
LTE	Long Term Evolution
LVDS	Low-Voltage Differential Signaling
MAC	Media Access Control
MCP	Multi-Chip Package
MH	Midhaul
MIG	Memory Interface Generator
MII	Media-Independent interface
MIMO	Multiple Input Multiple Output

MU-MIMO	Multiple User MIMO
NEBS	Network Equipment-Building System
NetConf	Network Configuration Protocol
NFV	Network Functions Virtualization
NIC	Network Interface Controller
NR	New Radio
O-CU	O-RAN Centralized Unit as defined by O-RAN
O-DU$_x$	A specific O-RAN Distributed Unit having fronthaul split option x where x may be 6, 7-2x (as defined by WG4) or 8
O-RU$_x$	A specific O-RAN Radio Unit having fronthaul split option x, where x is 6, 7-2x (as defined by WG4) or 8, and which is used in a configuration where the fronthaul interface is the same at the O-DUx
OCXO	Oven Controlled Crystal Oscillator
ORx	Observation Receiver
PAM	Power Amplifier Module
PCIe	Peripheral Component Interconnect express
PDCP	Packet Data Convergence Protocol
pFIR	Programable Finite Impulse Response filter
PHY	Physical Layer
PIM	Passive Intermodulation
PMBus	Power Management Bus
POE	Power over Ethernet
PPS	Pulse Per Second
PRACH	Physical Random-Access Channel
QAM	Quadrature Amplitude Modulation
QPSK	Quadrature Phase Shift Keying
QSFP	Quad Small Form-factor Pluggable
RAN	Radio Access Network

RDIMM	Registered Dual In-line Memory Module
RF	Radio Frequency
RFIC	Radio Frequency Integrated Circuit
RoE	Radio over Ethernet
RU	Radio Unit as defined by 3GPP
RX	Receiver
SATA	Serial ATA
SDU	Service Data Unit
SFP	Small Form-factor Pluggable
SFP+	Small Form-factor Pluggable plus
SOC	System On Chip
SPI	Serial Peripheral Interface
SSD	Solid State Drive
TCXO	Temperature Compensated Crystal Oscillator
TDP	Thermal Design Power
TR	Technical Report
TS	Technical Specification
TX	Transmitter
UL	Uplink
USB	Universal Serial Bus
WG	Working Group

Chapter 2 Hardware Reference Design 1

This chapter describes one example of white box hardware reference design including O-DU, O-RU.

2.1 O-CU Hardware Reference Design

The O-CU white box hardware is the platform that performs the O-CU function of upper L2 and L3. The hardware systems specified in this document meet the computing, power and environmental requirements of use cases configurations and feature sets of RAN physical node. These requirements are described in the hardware requirement specification in addition to the deployment scenarios document. The O-CU hardware includes the chassis platform, mother board, peripheral devices, and cooling devices. The mother board contains processing unit, memory, the internal I/O interfaces, and external connection ports. The midhaul (MH) and backhaul (BH) interface are used to carry the traffic between O-CU and O-DU as well as O-CU and core network. The other hardware functional components include: the storage for software, hardware and system debugging interfaces, board management controller, just to name a few; the O-CU designer will make decision based on the specific needs of the implementation.

The HW reference design of O-CU is the same as O-DU except for the need of a HW accelerator, thus the detail design will be described in O-DU section 2.2.

2.1.1 O-CU High-Level Functional Block Diagram

[Intentionally left blank. Refer to O-DU Functional Block Diagram]

2.1.2 O-CU Hardware Components

2.1.2.1 Digital Processing Unit

2.1.2.2 Hardware Accelerator (If required by design)

2.1.2.2.1 Accelerator Design 1

2.1.2.2.2 Accelerator Design 2

2.1.3 Synchronization and Timing

2.1.4 External Interface Ports

2.1.5 O-CU Firmware (if needed)

2.1.6 Mechanical

2.1.7 Power Unit

2.1.8 Thermal

2.1.9 Environmental and Regulations

2.2 O-DU Hardware Reference Design

The O-DU white box hardware is the platform that performs the O- DU functions such as upper L1 and lower L2 functions. The hardware systems specified in this document meet the computing, power and environmental requirements of use case's configurations and feature sets of RAN physical node. These requirements are described in the early hardware requirement specification as well as in the use cases document. The O-DU hardware includes the chassis platform, mother board, peripheral devices, and cooling devices. The mother board contains processing unit, memory, the internal I/O interfaces, and external connection ports. The fronthaul and backhaul interface are used to carry the traffic between O-RU/FHGW$_{7-2}$/FHGW$_{7-2\rightarrow 8}$ and O-DU as well as O-CU and O-DU. The O-DU design may also provide an interface for hardware accelerator if that option is preferred. The other hardware functional components include: the storage for software, hardware and system debugging interfaces, board management controller, just to name a few; the O-DU designer will make decision based on the specific needs of the implementation.

Note that the O-DU HW reference design is also feasible for O-CU and integrated O-CU/ O-DU.

2.2.1 O-DU High-Level Functional Block Diagram

In the following O-DU/O-CU mother board functional block diagram, it shows the interconnections of the major components and external interfaces. The digital processing unit or SoC (CPU) handles the baseband processing workload. To make the processing more efficient, an accelerator can be used to assist with the baseband workload processing. The memory devices include random-access memory (RAM) for temporary storage of data while flash memory is used for codes and logs. The storage device is for persistent storage. External and internal network cards can be used for fronthaul or backhaul connection. The baseboard management controller (BMC) is a microcontroller which monitors hardware operation on the motherboard. The clock circuits provide digital processing unit with required clock signals.

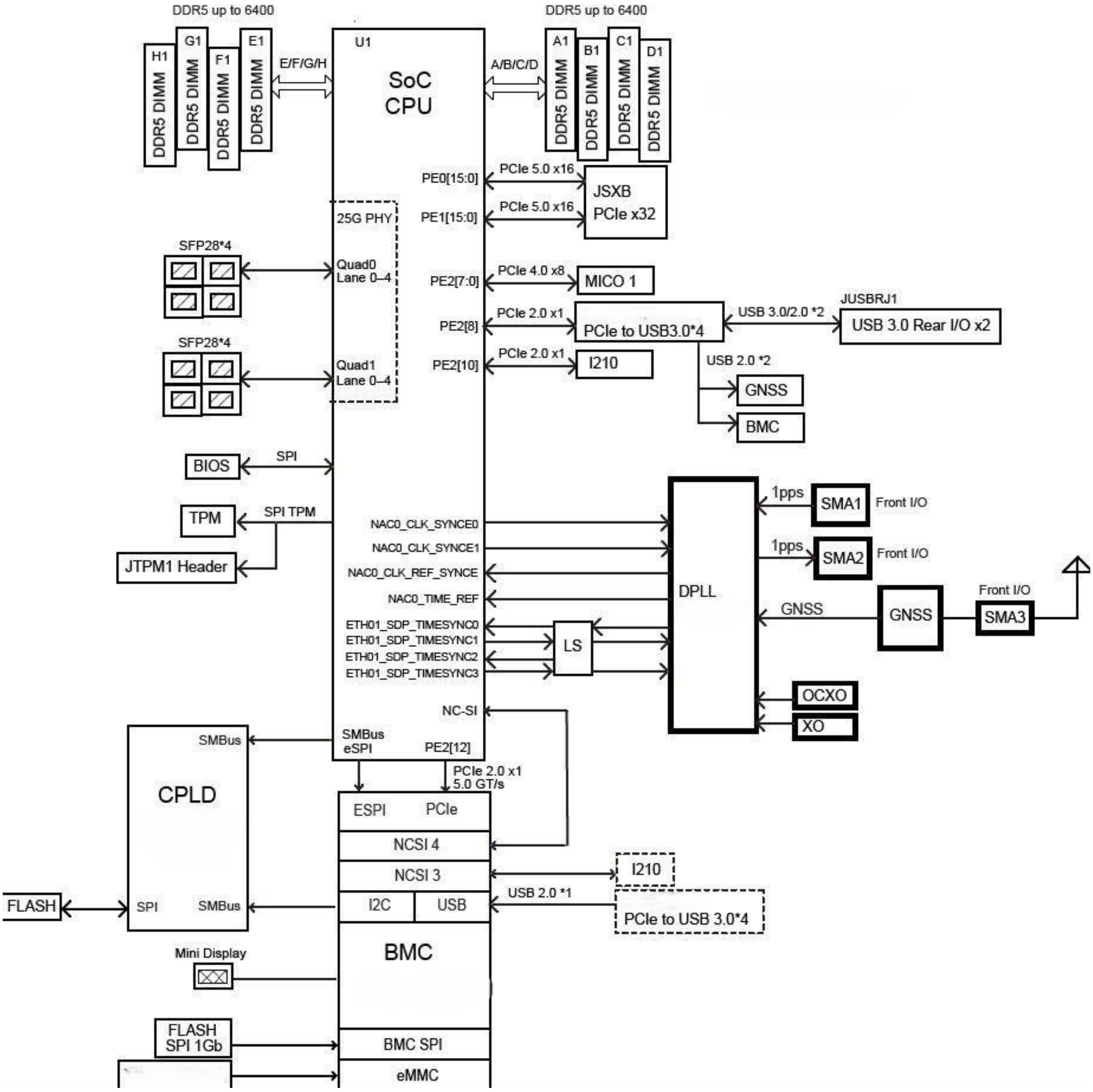


Figure 2.2.1-1 O-DU/O-CU Functional Block Diagram

2.2.2 O-DU Hardware Components

2.2.2.1 Digital Processing Unit

The example of general-purpose processor performances and other related information are listed in the following table.

Table 2.2.2-1 Processor Feature List

Item Name	Description
# of Cores	42
# of Threads	84
Base Frequency	2.30GHZ
Max Turbo Frequency	2.90GHz
Cache	168MB
TDP	235W
Max Memory Size (dependent on memory type)	512GB
Memory Types	DDR5 RDIMM
Max # of Memory Channels	4

2.2.2.1.1 Interfaces

These are the required interface specification for the main board.

2.2.2.1.1.1 Memory Channel Interfaces

The system memory capacity, type and related information is described in the following table.

Table 2.2.2-2 Memory Channel Feature List

Item Name	Description
Memory Types	DDR5
# of Memory Channels	4
ECC RDIMM\LRDIMM	Up to 512GB
3DS ECCRDIMM	Not supported
Memory Speed	up to 6400MT/s
DIMM Sizes	16GB, 32GB, 48GB, 96GB, 128GB
Memory Voltage	1.1 V

2.2.2.1.1.2 PCIe

PCIe Gen 5 is supported by the processor. There is a total of 48 PCIe lanes available on the motherboard which are allocated as per Figure 2.2.1-1 O-DU/O-CU Functional Block Diagram. This example has 2 x PCIe Gen5 x 16 slots. Each 16-lane bi-directional PCIe expansion slot has a total of 128GB/s of available bandwidth. There is also 1 PCIe 4.0 x8 via MCIO connector on the motherboard for further expansion options.

2.2.2.1.1.3 Ethernet

The system should be capable of offering a minimum aggregated 48 Gbps Ethernet bandwidth. The breakdown of the Ethernet ports is discussed in a later section.

2.2.2.2 Hardware Accelerator (if required by design)

2.2.2.2.1 Hardware Accelerator

Hardware accelerators can be used in O-DU to offload computationally intensive functions and to optimize the performance under varying traffic and loading conditions. While the hardware acceleration functional requirements and implementation are system designer's choice; the O-DU is required to meet the minimum system performance requirements under various loading conditions and deployment scenarios. In most cases, a Field Programmable Gate Array (FPGA) or Application Specific Integrated Circuit (ASIC) based PCIe card can be used to optimize the system performance. FPGA(s) and ASIC(s) may be part of a Network Interface Controller (NIC) that further provides connectivity services.

2.2.2.2.2 Accelerator Design 1

The O-DU system is typically implemented using a multi-core processor and one or more hardware accelerators. Parts of O-DU protocol stack can be implemented in software running on the multi-core processors and the computationally intensive L1 and L2 functions can be offloaded to an FPGA-based hardware accelerator. This accelerator comprises two FPGAs for L1 offload and fronthaul connectivity in lookaside mode. The accelerator hardware and firmware features are listed in Table 2.2.2-3 and Table 2.2.2-4.

Table 2.2.2-3 Accelerator Hardware Features

SoC Resources	**FPGA1**	**FPGA2**
	System Logic cells - 930K CLB LUT - 425K SDFEC -8 DSP Slices - 4,272 BRAM - 38.0Mb URAM - 22.5Mb	System Logic cells - 1,143K CLB LUT - 523K CLB Flip-Flops -1,045K DSP Slices - 1,968 BRAM - 34.6Mb URAM - 36.0Mb
Form Factor	FHHL PCIe Form Factor	
PCIe Interface	Gen 3 x16 with Bifurcation	
On Board Memory	**FPGA1**	**FPGA2**
	Total Capacity 4 GB in PL, upgradeable to 8GB Total Capacity 2 GB in PS, upgradeable to 4GB	Total Capacity 4 GB in PL, upgradeable to 8GB Total Capacity 2 GB in PS, upgradeable to 4GB
Network Interface(s)	2xSFP28 optical interfaces to FPGA2 (User Configurable, includes 10/25 Ethernet)	
Other External Interface(s)	Micro USB for JTAG support (FPGA programming and debug) and access to BMC	

Graphical User interface	GUI for monitoring the basic board parameters, monitoring temperature alerts, firmware upgrades for BMC
Board Management Controller	Telemetry, Security, Remote Upgrade
Clocking	O-RAN C1, C2, C3, C4
Fronthaul	eCPRI, RoE IEEE1914.3, O-RAN WG4 FH
GPS	SMA for 1 PPS (In/Out)
Operating Temperature	NEBS Compliant
Power	< 75 W
Clocking Options	Low-Jitter, configurable clock ranging from10MHz to 750MHz 1 PPS input and output with assembly option for OCXO and TCXO

Table 2.2.2-4 Hardware Accelerator Firmware Features

Item Name	Description
Remote System Upgrade	Securely upgradable FPGA flash image
L1 Acceleration	5G NR LDPC encoding/decoding with interleaving/de-interleaving and rate-matching/rate-de-matching along with early termination, CRC attachment, and HARQ management
Hardware-Software APIs	Fully compliant with DPDK/BBDEV APIs
Fronthaul Protocols/Compression	Fully compliant with O-RAN WG4 MCUS-planes Specifications
Open Programmable Acceleration Environment	Support for: • FPGA flashing upgrade • Firmware version reporting • PCIe diagnostics • Ethernet diagnostics • Temperature and voltage telemetry information

The hardware accelerator supports x86 or non-x86-based processors. Figure 2.2.2-1 illustrates the two-chip acceleration architecture comprising two FPGAs with multi-lane PCIe interfaces toward the CPU and external connectivity toward O-RU(s) via eCPRI and O-CU(s) through GbE connectivity. The example architecture further depicts multi-lane Gen3 PCIe interfaces between each FPGA and the CPU. The FPGAs communicate through high-bandwidth Ethernet (GbE) transport.

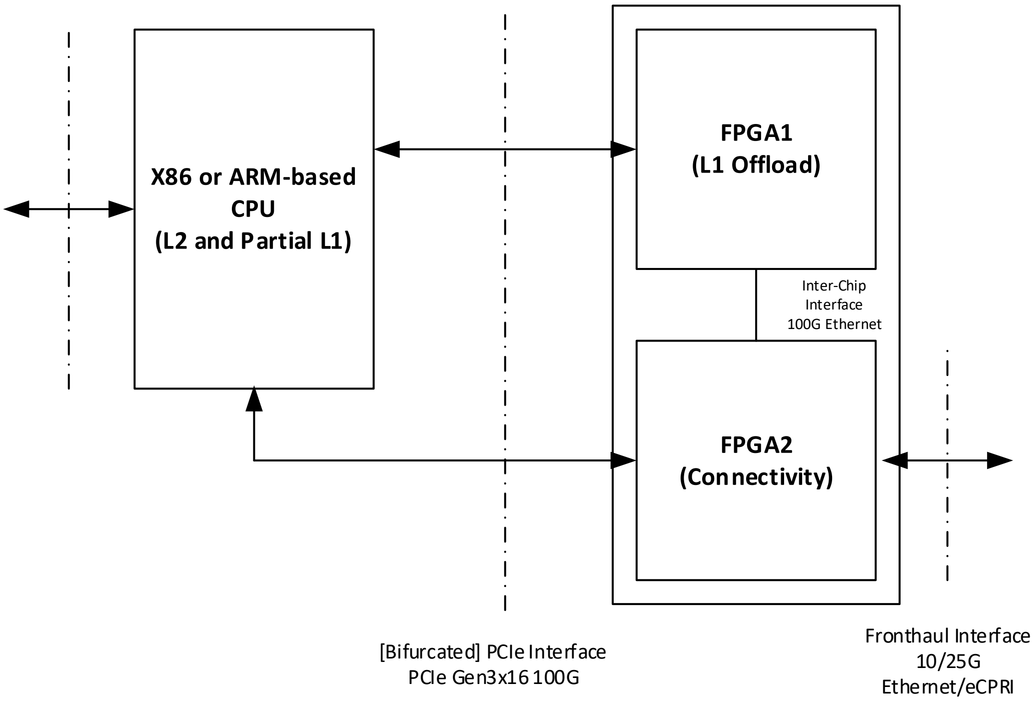


Figure 2.2.2-1 Dual-chip FPGA-based Hardware Acceleration in O-DU

2.2.2.2.3 Accelerator Design 2

The O-DU system is typically implemented using a multi-core processor (CPU) and one or more hardware accelerators. Parts of O-DU protocol stack can be implemented in software running on the multi-core processors and the computationally intensive L1 and L2 functions can be offloaded to an FPGA-based hardware accelerator. This accelerator comprises a single FPGA for L1 offload in lookaside mode. The accelerator hardware and firmware features are listed in Table 2.2.2-5 and Table 2.2.2-6.

Table 2.2.2-5 Accelerator Hardware Features

SoC Resources	System Logic cells - 930K CLB LUT - 425K SDFEC -8 DSP Slices - 4,272 BRAM - 38.0Mb URAM - 22.5Mb
Form Factor	HHHL PCIe Form Factor
PCIe Interface	Gen 3 x16 (Gen4 x8)
On Board Memory	Total Capacity 4 GB in PL, upgradeable to 8GB Total Capacity 2 GB in PS, upgradeable to 4GB
Other External Interface(s)	Micro USB for JTAG support (FPGA programming and debug) and access to BMC
Graphical User interface	GUI for monitoring the basic board parameters, monitoring temperature alerts, firmware upgrades for BMC

Board Management Controller	Telemetry, Security, Remote Upgrade
Operating Temperature	NEBS Compliant
Power	< 35 W

Table 2.2.2-6 Hardware Accelerator Firmware Features

Remote System Upgrade	Securely upgradable FPGA flash image
L1 Acceleration	5G NR LDPC encoding/decoding with interleaving/de-interleaving and rate-matching/rate-de-matching along with early termination, CRC attachment, and HARQ management
Hardware-Software APIs	Fully compliant with DPDK/BBDEV APIs
Open Programmable Acceleration Environment	Support for: • FPGA flashing upgrade • Firmware version reporting • PCIe diagnostics • Ethernet diagnostics • Temperature and voltage telemetry information

The hardware accelerator supports x86 or non-x86-based processors. Figure 2.2.2-2 illustrates the single-chip acceleration architecture comprising one FPGA with Gen3 x16 or Gen4 x8 PCIe interfaces toward the CPU.

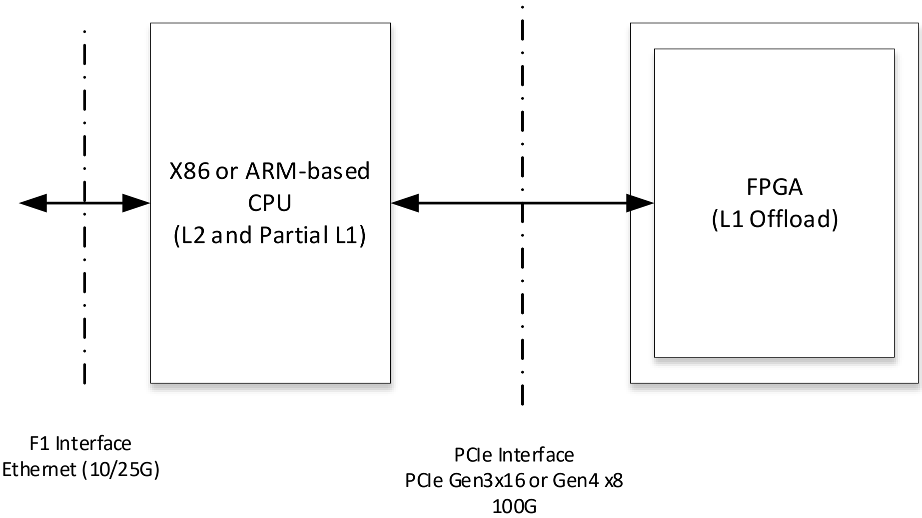


Figure 2.2.2-2 Single-chip FPGA-based Hardware Acceleration in O-DU

2.2.2.2.4 Accelerator Design 3

This design includes an Embedded NIC Inline Accelerator (ENIAC) with an ASIC that can provide NIC functions. One option supports a high-capacity O-DU using multiple PCIe cards in standard racks and aligns with the FAPI interface to maximize software reuse. The second option supports a high-capacity O-DU using an Ethernet-based backplane interconnect and aligns with the nFAPI interface to maximize software reuse. ENIAC unloads L1 from the host, allowing it to focus on L2.

a. Accelerator Requirements

Hardware and firmware requirements for this accelerator design are given in Table 2.2.2-7 and Table 2.2.2-8, respectively.

Table 2.2.2-7 Accelerator Hardware Feature List

Item Name	Description
PCIe (Interface with digital processing unit)	Gen4 x16 (and lower)
Form factor	FHHL
Connectivity	QSFP28/SFP
NIC Device	200Gb xHAUL for FH, BH & MH traffic shaping.
DDR Main	TBD GB DDR4
Flash (FPGA images)	>=1 Gbit
BMC	Telemetry, Security, remote upgrade
Clocking	For O-RAN C1, C2, C3 & C4
Fronthaul	eCPRI, RoE IEEE1914.3, O-RAN WG4
GPS	SMA for 1 PPS & 10MHz (in/out)
Operating Temperature (ambient)	NEBS Compliant
Power	<75W
Clock Accuracy	Low-Jitter, configurable clock ranging from 10MHz to 750MHz. Option for OCXO (TCXO as standard)

Table 2.2.2-8 Accelerator Firmware Feature List

Item Name	Description
Remote system upgrade	Securely upgradable image
Queuing	Command queuing at FAPI interface
LDPC Acceleration	Rel15 compliant
Fronthaul Compression	In-line compression/decompression for Mu-Law, block-floating point, and quantization according to the O-RAN WG4 specification.
Open programmable acceleration environment	Support for: • Flashing upgrade • Firmware version reporting • PCIe diagnostics • Ethernet diagnostics • Temperature and voltage telemetry information.

b. Accelerator Design

Figure 2.2.2-3 illustrates Accelerator Design 3 with integrated NIC and PCIe connectivity to L2.
Figure 2.2.2-4 illustrates Accelerator Design 3 with integrated NIC and Ethernet connectivity to L2.

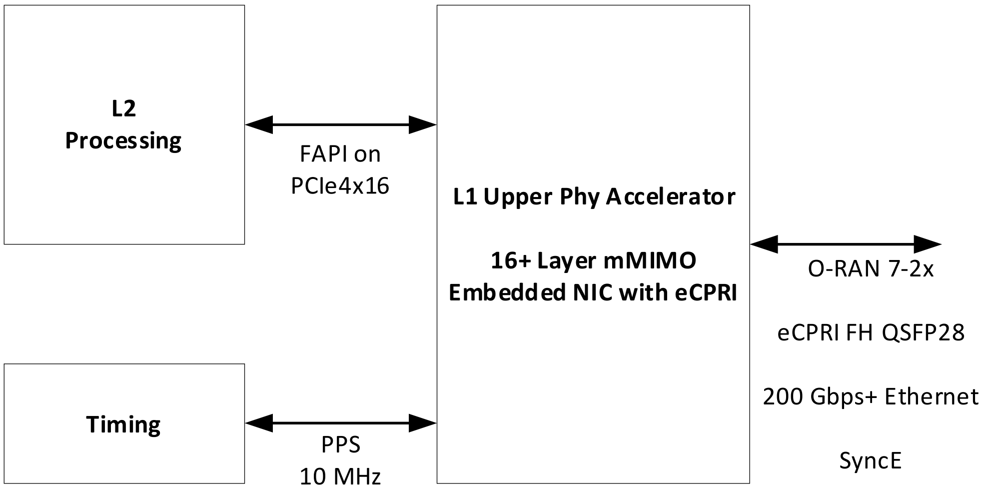


Figure 2.2.2-3 Accelerator Design 3 using PCIe

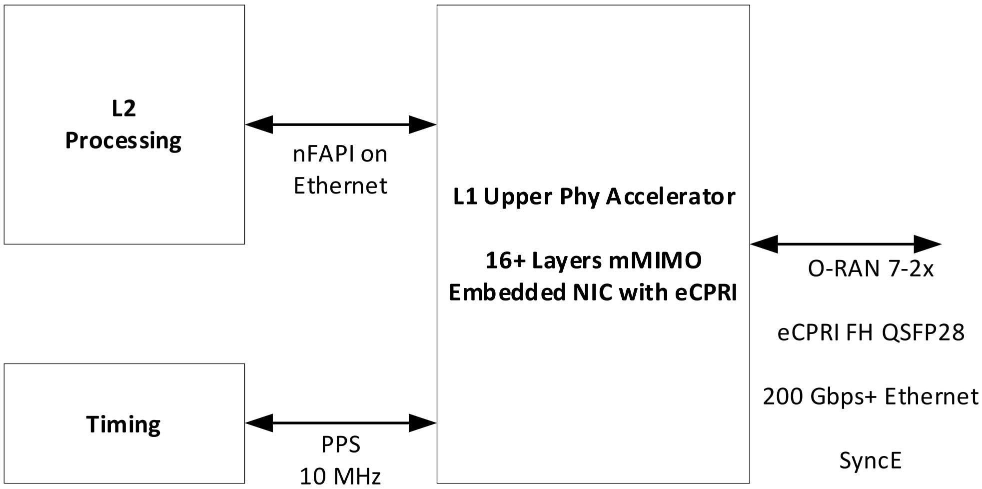


Figure 2.2.2-4 Accelerator Design 3 using Ethernet

2.2.3 Synchronization and Timing

In this reference design the DU has a GNSS receiver integrated on the motherboard. The GNSS receiver is connected via SMA cable to an external GNSS antenna.

The PTP and SyncE signals are transported by Ethernet L2 via the fronthaul network interfaces to the O-RU. Together these provide the local time of day and frequency reference for the O-RU.

Summary of timing-related software features supported:

- PTP distribution compliant to G.8272
- ITU-T specifications for SyncE:
  - Definitions: ITU-T G.8260
  - Architecture: ITU-T G.8261
  - SSM transport channel and format: ITU-T G.8264
  - Clock specifications: ITU-T G.8262 (EEC)
  - Functional model and SSM processing: ITU-T G.781
- IEEE 1588 Telecom Grand Master (T-TGM) using G8275.1 profile.
- T-GM conversion to Telecom Boundary Clock (T-TBC) if GNSS signal is lost.
- Telecom Boundary Clock (T-TBC) per ITU-T G.8273.2.
- Min 4-hour holdover/drift of max 0.65 µsec, temperature change of +5°C or -5°C after Holdover starts.
- Compliant with input and output jitter and wander requirements specified in ITU-T G.8262 (for EEC).
- Alternate BMCA specified in G.8275.1.

2.2.4 External Interface Ports

The following table shows the external ports or slots that the system provided. In addition to a single 1GbE management port, the SoC provides 8 additional 25G SFP Ethernet ports. In this configuration the system has 200Gbps of aggregated Ethernet bandwidth. A further 8 x 25G SFP Ethernet ports can be installed via PCIe expansion slot for a total of 16 x 25G or 400Gbps aggregated Ethernet bandwidth.

Table 2.2.4-1 External Port List

Port Name	Feature Description
Ethernet	1 x GbE BMC management port
	8 x 25G SFP
PCIe	2 x PCIe 5.0 x16 FHFL slots
	1 X PCIe 4.0 x 16 MCIO connector
USB	2 x USB 3.0 ports
VGA	VGA port
Serial Port	COM port via RJ45
SMA	GNSS Antenna in
	10Mhz out
	1PPS out

2.2.5 O-DU Firmware

BIOS and BMC firmware are needed in the system and shall be installed.

2.2.6 Mechanical

2.2.6.1 Mother Board

The mechanical layout of the mother board is detailed in Figure 2.2.6-1 Mother Board Layout Diagram and shows the location of major components and interface ports.

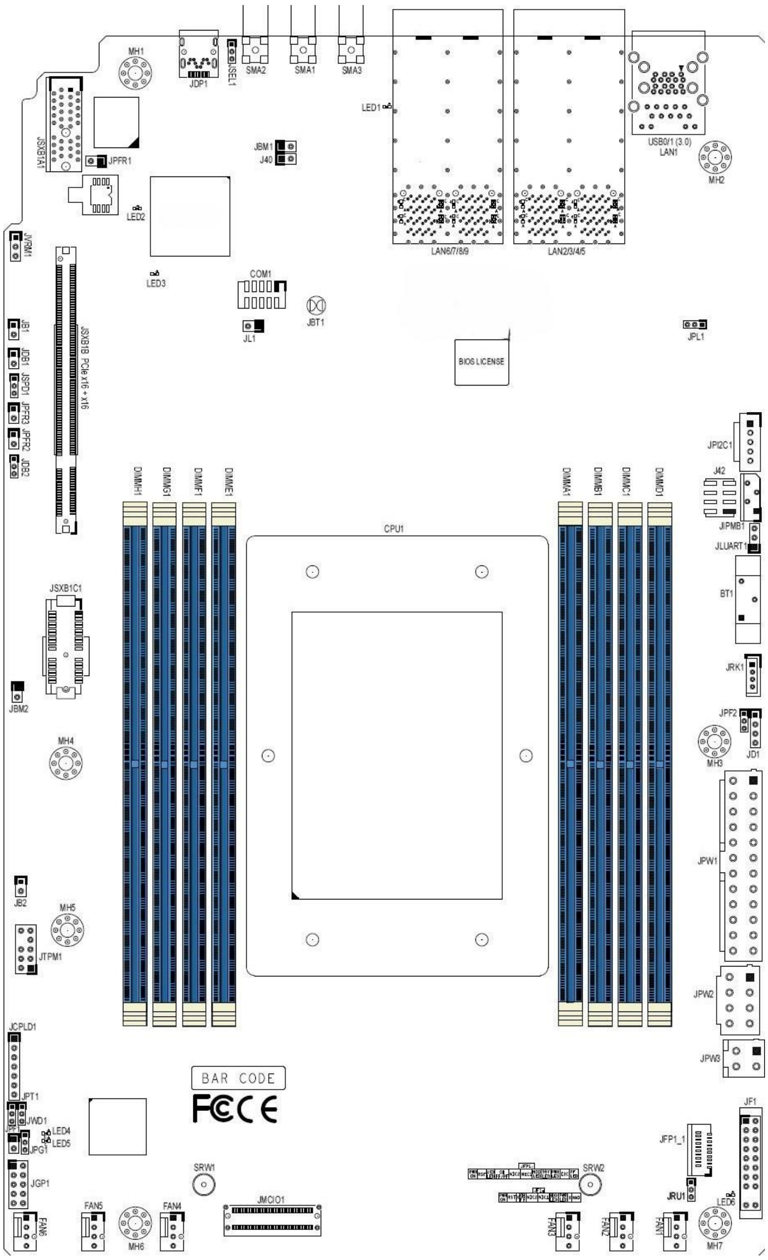


Figure 2.2.6-1 Mother Board Layout Diagram

2.2.6.2 Chassis

The 1U rack mount chassis contains the layout of the power supply, SSD, and fans. The chassis dimensions and layout are detailed in Figure 2.2.6-2 Chassis Mechanical Diagram.

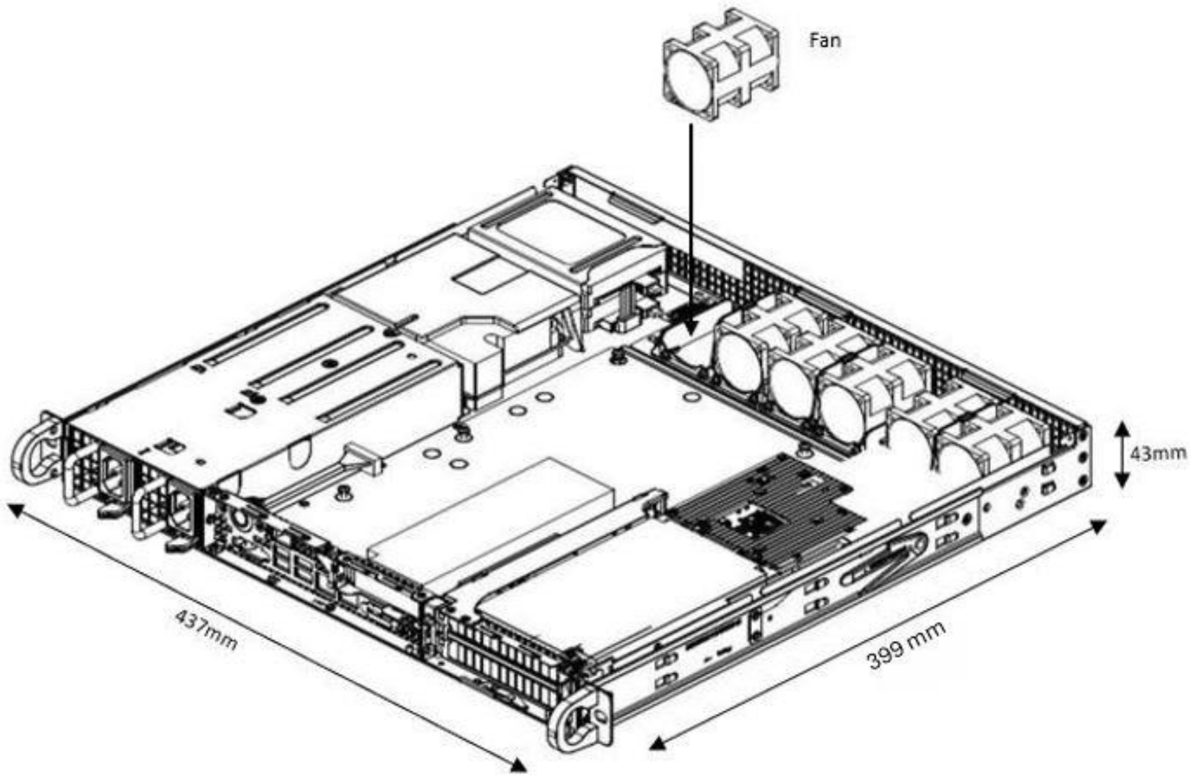


Figure 2.2.6-2 Chassis Mechanical Diagram

2.2.6.3 Cooling

The system is supplied with 6 x 40x56mm PWM fans for cooling.

2.2.7 Power Unit

In a fully loaded system with two PCIe slots populated, the system power consumption should be less than 400W. The total system power shall be kept at less than 80% of the power supply capacity.

2.2.7.1 Power Supply

Power is provided by dual redundant 600W DC 48V High-Efficiency power supplies w/PMBus 1.2. The power support input and output power rails are listed below.

Table 2.2.7-1 O-DU Power Requirements

Supply Domain	Requirements
DC Input	-40Vdc to -65Vdc
DC Output	+5Vsb: 3A
	+12V: 50A

2.2.8 Thermal

Active cooling with up to 6 fans is integrated in the chassis.

The hardware acceleration cards described in Section 2.2.2.2 use passive cooling and a custom heatsink and are equipped with temperature sensors. It is designed to operate in temperatures ranging from -5°C to +55°C.

2.2.9 Environmental and Regulations

The O-DU hardware system is RoHS Compliant. The power supply unit is EMC FCC/CISPR Class B compliant. Table 2.2.9-1 lists the environmental related features and parameters.

Table 2.2.9-1 Environmental Features

Item Name	Description
Operating Temperature	-5°C to +55°C
Non-operating Temperature	-40°C to 70°C
Operating Relative Humidity	8% to 90% (non-condensing)
Non-operating Relative Humidity	5% to 95% (non-condensing)

The hardware accelerator described in Section 2.2.2.2 is designed to operate in indoor environments and in temperatures ranging from -5°C to +55°C.

2.3 O- RU Hardware Reference Design

This chapter defines the hardware reference design for the Outdoor Macrocell O-RU. This is defined by a 7-2 interface to the O-DU on one side and up to 64T64R on the antenna port (more details on the mMIMO O-RU$_{7\text{-}2x}$ reference designs are given in Section 2.5). No upper limit is defined to the output power of Wide Range BS according to 3GPP TS38.104-Section 6.2 [3], the external power is supplied by either AC or a standard DC -48V telecom source.

2.3.1 O- RU High-Level Functional Block Diagram

Figure 2.3.1-1 provides a high-level functional block diagram depicting the major HW/SW components. It also highlights the internal/external interfaces that are required. This document shows how to implement the system defined by the OMAC-HAR [6] document, and mMIMO O-RU$_{7\text{-}2x}$, which can be realized with different high level radio architecture based on various functional splits and interface/interconnects between these functional blocks.

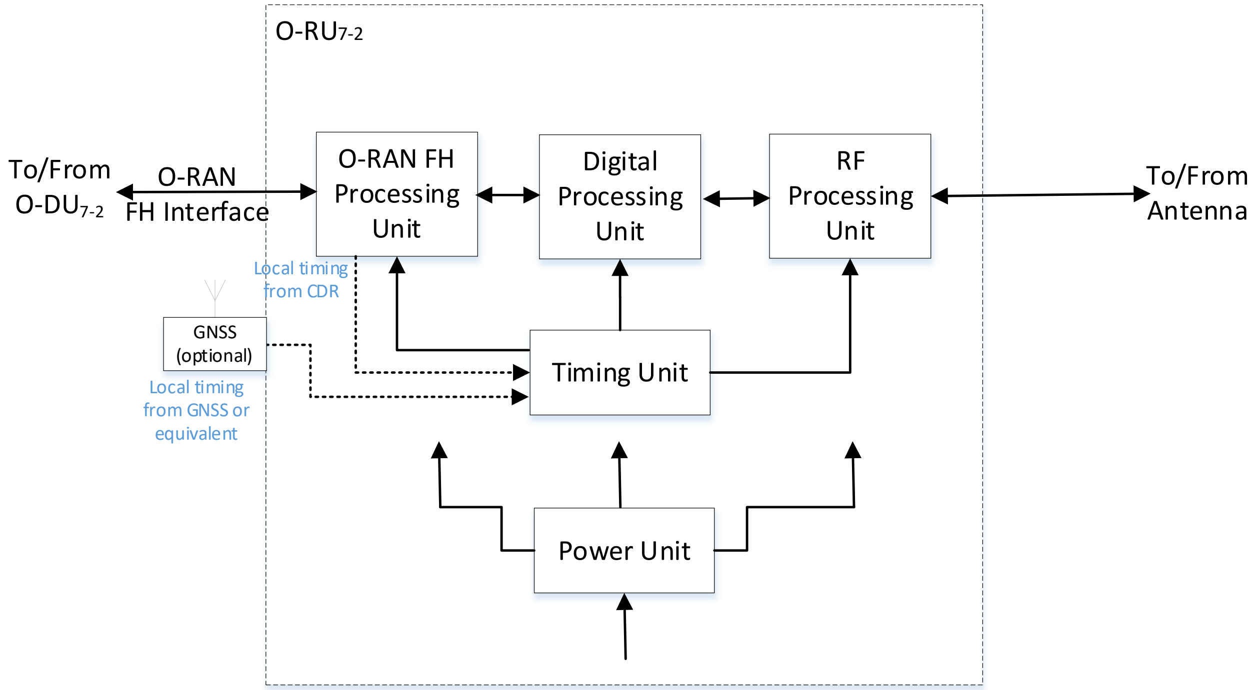


Figure 2.3.1-1 O-RU High-Level Functional Block Diagram

2.3.2 O-RU Hardware Components

Figure 2.3.2-1 shows the high-level block details of O-RU Hardware implementation example. The description of each block from left to right is given below.

O-RU$_{7\text{-}2x}$ communicates with O-DU$_{7\text{-}2x}$ through O-RAN Fronthaul interface and the control/data packets are processed by O-RAN Fronthaul unit with the help of CUS plane processing block. Subsequently the packets are processed by Low-PHY functional blocks such as encoding, scrambling, modulation, layer mapping, precoding, beamforming, and resource element mapping in downlink. The reverse operation is done during the uplink process. The output of the Low-PHY block is fed to digital processing unit through packet interface. Here the frequency conversion in digital domain i.e., digital up conversion (DUC) in DL and digital down conversion (DDC) in UL are done. Furthermore, DPD and CFR algorithms are implemented in digital processing unit to enhance the power amplifier efficiency by reducing the PAPR/ACLR in RF front end. The output of digital processing unit is fed to RF processing unit through digital interface. The digital signals converted to analog in DL and vice versa in UL with the help of DAC and ADC functional blocks in Transceiver unit, respectively. Signals are further amplified with the help of functional blocks like PA in DL and LNA in UL in analog RF front end unit. The output of the amplifier is fed to a cavity filter to suppress the unwanted signals in DL and is done before amplification in the case of UL operation i.e., receive operation. The TDD switch is used to switch the operation between DL and UL, i.e., transmit and receive operations. The output of the cavity filter i.e., the O-RU is fed to antenna unit and then transmitted over the air. Three processing units such as FH, Digital, and RF powered up using the power unit and gets synchronized with the help of timing/synchronization block. In the case of beamforming scenario, antenna calibration is implemented using antenna calibration processing block. This coordinates with Low-PHY processing unit and calibration switch (Cal switch) and processes the signals coming from and going to the antenna calibration port, to compensate for the amplitude and phase offsets existing in each RF chain. Other implementations are also possible by rearranging the functional blocks of O-RAN FH, Digital processing, and RF processing units.

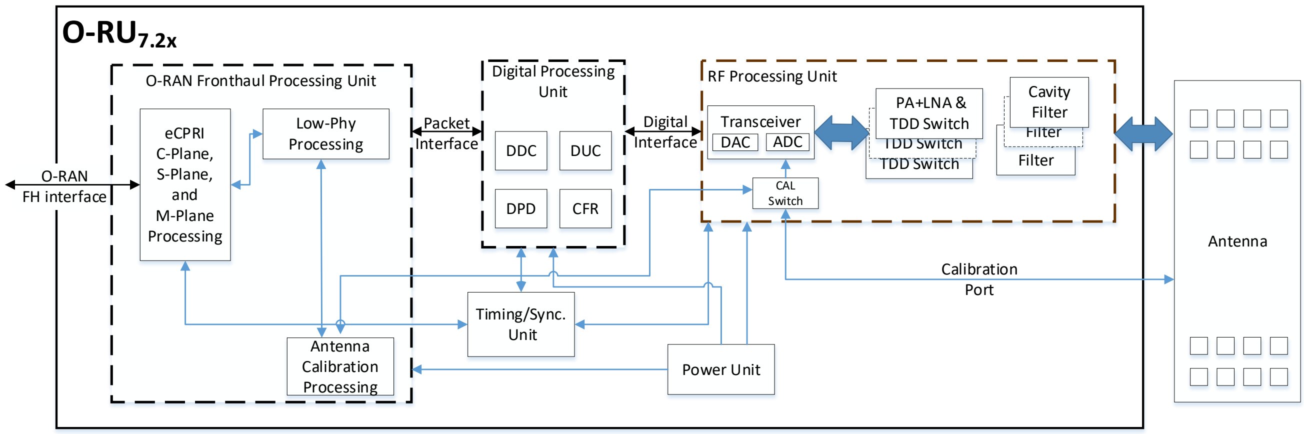


Figure 2.3.2-1 O-RU Architecture Example Implementation 1

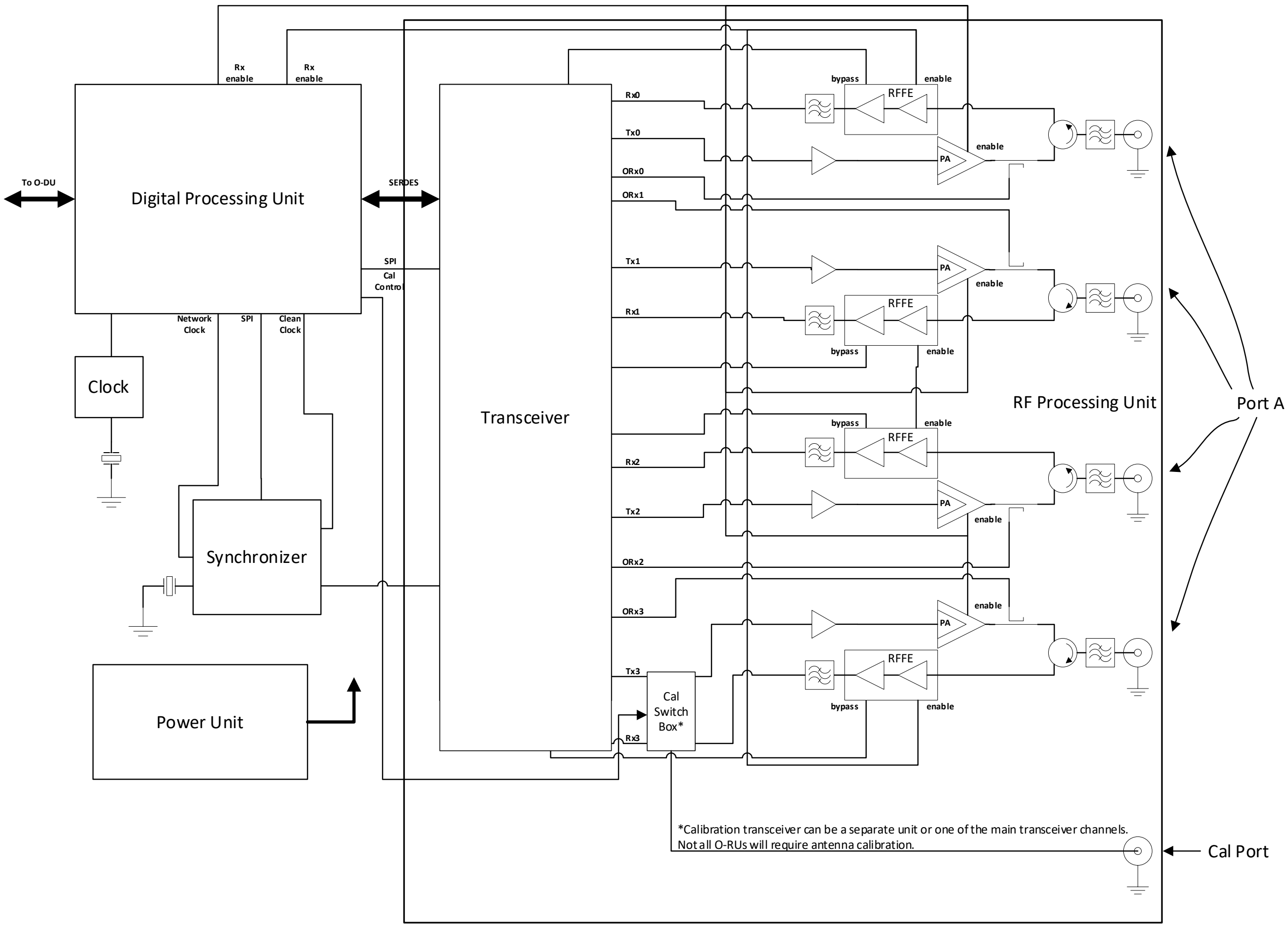


Figure 2.3.2-2 4T4R General Block Diagram with full RF front end

Figure 2.3.2-2 and Figure 2.3.2-3 shows an example of an O-RU design for 4T4R and 8T8R, respectively. In subsequent sub-sections, we present detailed hardware components the O-RAN Processing Unit and Digital Processing Unit, RF processing unit, Power unit and Clock/Synchronization units.

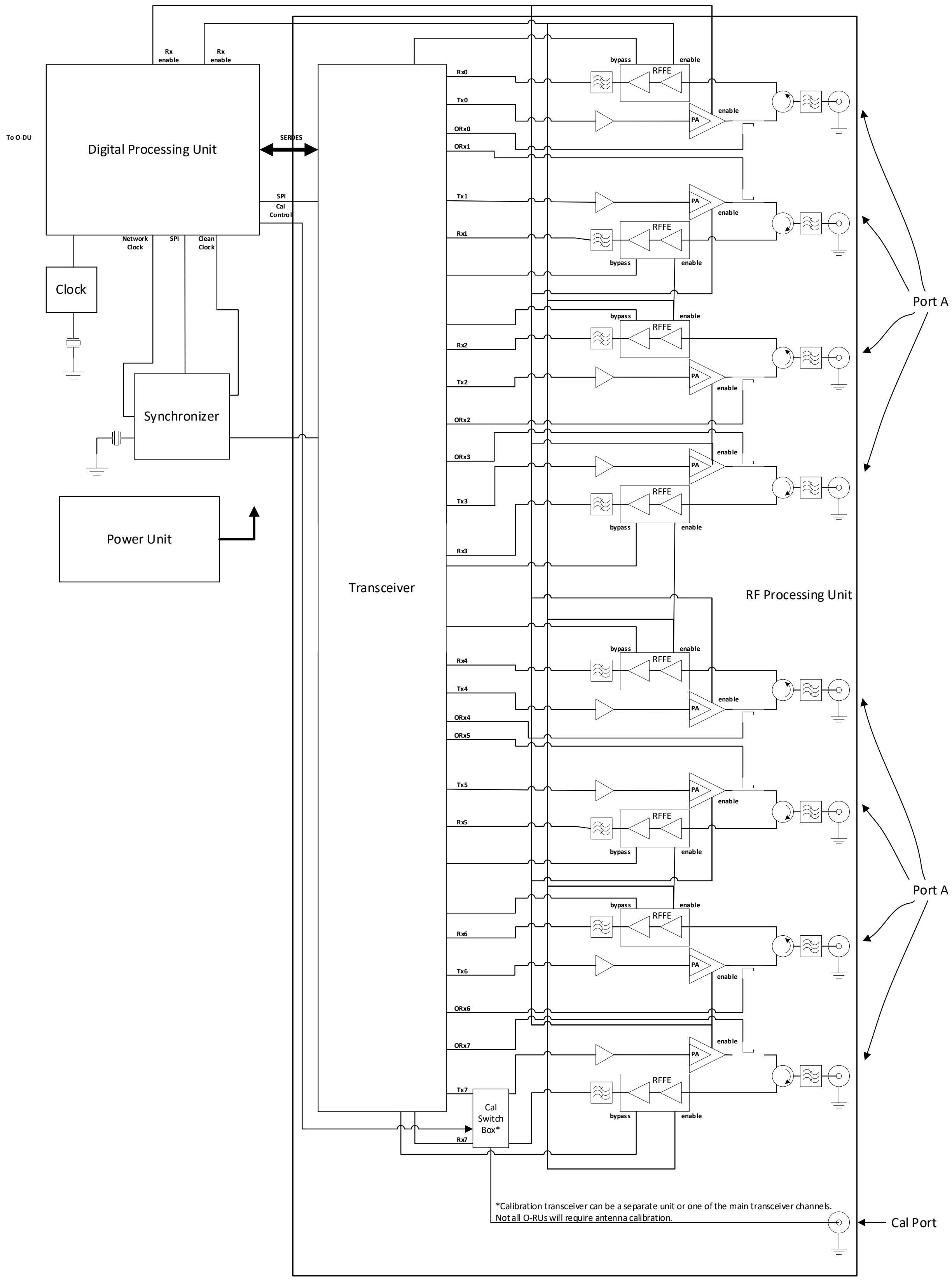


Figure 2.3.2-3 8T8R General Block Diagram with RF front end

2.3.2.1 O-RAN Processing Unit

The O-RAN Processing Unit (OPU) receives eCPRI frames from the O-RAN fronthaul and performs the fronthaul interface, lower L1, synchronization and beamforming.

2.3.2.2 Digital Processing Unit

The Digital Processing Unit (DPU) of the O-RU performs synchronization, DDC, DUC, CFR, and DPD. The DPU can be implemented as a FPGA or ASIC.

The following block diagrams describe the digital processing unit as per example implementation in Figure 2.3.2-4 and Figure 2.3.2-5.

Other implementations are possible.

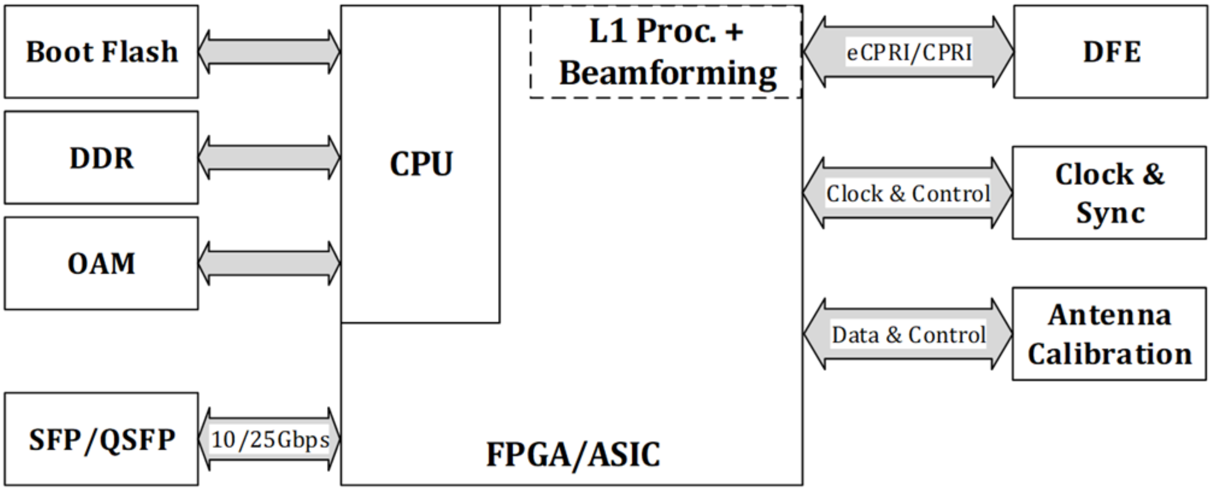


Figure 2.3.2-4 Digital Processing Unit Block Diagram (L1 Processing + Beamforming)

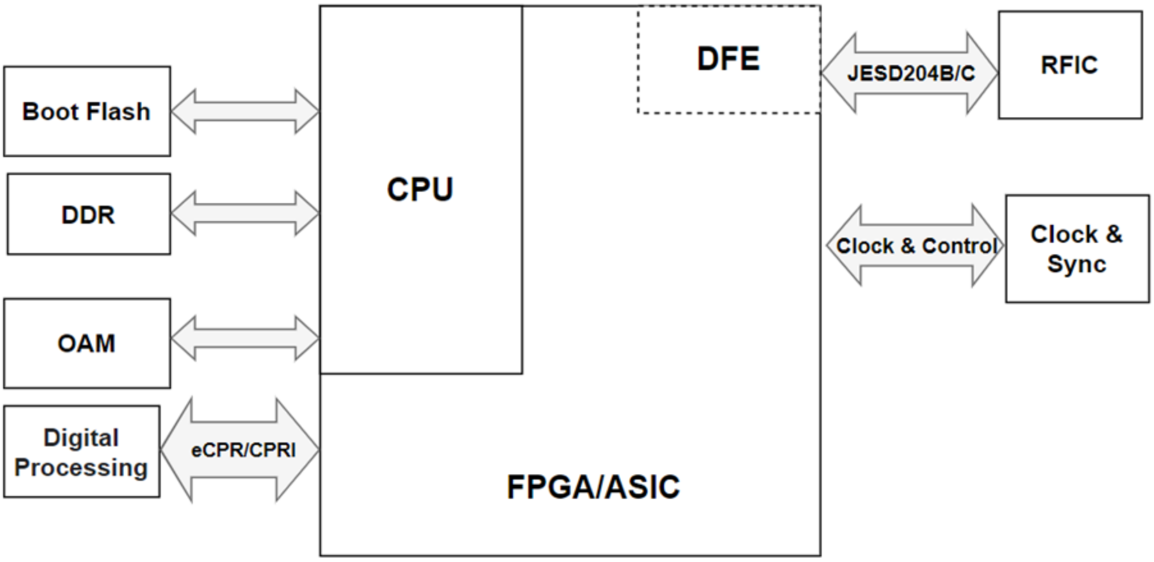


Figure 2.3.2-5 Digital Processing Unit Block Diagram (DFE)

2.3.2.2.1 FPGA/ASIC Solution

This solution for the digital processing unit is based on an FPGA with embedded ARM processor.

2.3.2.2.1.1 FPGA Requirements

Table 2.3.2-1 shows the general interfaces required for an FPGA based design based on Figure 2.3.2-4 and Figure 2.3.2-5.

Table 2.3.2-1 FPGA Interface Requirements

Item Name	Description
eCPRI Port	High speed bi-directional interface transporting data and control information. Total payload will be determined by the antennas/layers served, bandwidth supported, bit precision, subcarrier spacing, etc. It is typically served by one or more QSFP+ or SFP28 connectors depending on the payload.
SERDES	Up to 16 high speed JESD204B/C serial interface lanes between FPGA fabric and transceiver depending on transceiver configuration selected.
Boot Clock	Clock for any uC or fabric functions required immediately at power on
Sysref	JESD204B/C synchronization signal
DevCLK	JESD204B/C device clock
ARM Clock	Clock for the ARM processor
Memory Clock	Memory clock
OAM Port	Control interface used for maintenance
DDR	Interface for external DDR memory

2.3.2.2.1.2 FPGA Design

Table 2.3.2-2 shows the general blocks required for the design. The core elements implemented in this design are required to be sized to support the requirements shown in the OMAC-HAR [6] document. Specifically, the DPU should support up to 200 MHz of occupied BW with up to two 100 MHz component carriers. In addition to signal processing, the DPU should support C, U, S and M plane handling. Major processing blocks in the DPU are shown in Table 2.3.2-2.

Table 2.3.2-2 FPGA Functional Blocks

Item Name	Description	Cat A	Cat B
FFT & CP-	FFT and Cyclic Prefix removal for uplink processing	Required	Required
iFFT & CP+	iFFT and Cyclic Prefix insertion for downlink processing	Required	Required
PRACH Front End	PRACH front end processor	Required	Required
Channel DDC	Component Carrier Digital Down Converter	Optional	Optional
Channel DUC	Component Carrier Digital Up Converter	Optional	Optional
CFR	Crest Factor Reduction	Optional	Optional
DPD	Digital Pre-Distortion	Optional	Optional
JESD204B/C	JEDEC High speed serial interface type B or C framer/de-framer	Required	Required
C/U/M Plane	Control, User, and management Plane processing	Required	Required
1588v2 Stack	IEEE 1588v2 Synchronization Software Stack	Required	Required
eCPRI	evolved Common Public Radio Interface framer/de-framer	Required	Required
I/Q Compression	I/Q compression/decompression for fronthaul lane rate reduction	Optional	Optional
Beamforming	Create & steer several beams	Optional	Optional
Pre-coding	Pre-coding	Optional	Required
RE Mapping	Resource Element Mapping to Antenna port	Optional	Required

2.3.2.3 RF Processing Unit

The RF Processing Unit of O-RU includes the transceiver block, up/down converters, power amplifiers (PAs), low noise amplifiers (LNAs), Tx/Rx filters. All conversions between analog and digital domain (e.g., (RF sampling, frequency conversion using mixing of RF, IF and LO in up and down conversion) are performed within the transceiver block. Note that physical and logical partitioning within the RF processing unit need not occur at any specific boundary.

2.3.2.3.1 Transceiver Reference Design

For the O-RU the sampling function and frequency conversion function is performed by transceiver. The purpose of the transceiver is to translate the RF signal to digital as early as possible in the signal chain and to reduce the data rate as low as possible to save power. The Transceiver is to convert between high-speed baseband data and a low-level RF for both transmit and receive signal chain. The transceiver may be responsible for orchestration of external gain control elements in both the receive and transmit paths. Other radio functions may need to be controlled by the DPU where they need to be controlled fast and synchronized with the TDD frame structure. Such elements include the transmit / receive switch, PA and LNA enable.

2.3.2.3.1.1 Hardware Specifications

2.3.2.3.1.1.1 Interface

The following table shows the detail of the RF Processing Unit Interface Specifications:

Table 2.3.2-3 RF Processing Unit Interface Specifications

Item Name	Description
High Speed Data (SERDES Out, SERDES In)	High speed data represents the time-domain baseband information being transmitted or received. Depending on the configuration of the O-RAN device, various bandwidths may be supported leading to a range of payload rates. Options for data include JESD204B and JESD204C, or proprietary formats. Several options are shown in the following table. All data represents IQ 16-bit (N=16) precision. Some devices support IQ 12 bit (N'=12) which may reduce the required data rates accordingly.
Reference Clock (REF Clock In)	The transceiver should receive a reference for internal clock and LO synthesis needs. This reference clock can function as the JESD204 Device Clock where the interface is by SERDES. The specific clock frequency is determined by the operation mode of the transceiver.
Control	Provides the control functionalities of the transceiver. As an example, SPI or I2C may be supported, or other proprietary solutions.
GPIO (GPIO 1 – GPIO #)	The transceiver may optionally include GPIO for controlling peripherals including but not limited to PAs, LNAs and other devices. These GPIOs should at a minimum support 1.8V outputs but the specifics will be determined by the connected devices. The GPIO should also support input from peripheral devices. Input should at a minimum support 1.8V logic with tolerance of 3.3V preferred. An example could be MIPI/ GRFC interfaces.
Tx Enable	The transceiver should provide an output to support enabling and disabling the external devices in the transmit chain such as a TxVGA (optional) and PA. This may be part of the GPIO pins.
Rx Enable	The transceiver should provide an output to support enabling and disabling the external devices in the transmit chain such as a RF Front End Module or LNA. This may be part of the GPIO pins.
LNA Bypass	The transceiver should provide an output to support bypassing the LNA appropriately in the condition of a large blocker if so required. This may be part of the GPIO pins.
RF Outputs (Tx1 – Tx4 or Tx8)	RF outputs including the main Tx signal should support 50 ohm or 100 ohms signalling. These outputs can be either single ended or differential.
RF Inputs (Rx1 – Rx4 or Rx8 and ORx1 – ORx4 or ORx8)	RF inputs including the main Rx and the Observation Rx (ORx) (for DPD) should support 50 ohm or 100 ohms signalling. These inputs can be either single ended or differential. The device should support at least 1 ORx.

Table 2.3.2-4 JESD204B/C Serial Data Rates

Bandwidth	JESD204B	JESD204C	JESD204B	JESD204C
	4T4R		8T8R	
20	4.9152	4.05504	9.8304	8.11008
50	9.8304	8.11008	19.6608	16.22016
100	19.6608	16.22016	39.3216	32.44032
200	39.3216	32.44032	78.6432	68.88064

2.3.2.3.1.1.2 Algorithm

The transceiver is required to provide appropriate algorithms to sustain RF operation including but not limited to Rx AGC, Tx Power control, DPD and CFR.

2.3.2.3.1.1.3 Device Configuration

The transceiver should support either 4T4R or 8T8R. In addition, at least one ORx path should be provided to support DPD functionality.

2.3.2.3.1.1.4 Power Dissipation

Total dissipation of transceiver IC for TRx should be less than 6 W for 4T4R.

2.3.2.3.1.1.5 RF Specifications

Table 2.3.2-5 Transceiver RF Specifications

Parameter	Symbol	Min	Typ	Max	Unit	Test Condition/Comment
Transmitters						
Centre Frequency		650		6000	MHz	
Transmitter Synthesis Bandwidth				450	MHz	
Transmitter Large Signal Bandwidth				200	MHz	
Transmitter Attenuator Power Control Range		0		32	dB	Signal to noise ratio (SNR) maintained for attenuation between 0 and 20 dB
Transmitter Attenuation Power Control Resolution			0.05		dB	20 MHz LTE/NR at -12 dBFS
Adjacent Channel Leakage Ratio (ACLR)			-66		dB	75 MHz <f≤2800 MHz
In Band Noise Floor			-154.5		dBm/Hz	0 dB attenuation; in band noise falls 1 dB for each dB of attenuation for attenuation between 0 dB and 20 dB
Out of Band Noise Floor			-153		dBm/Hz	600 MHz < f ≤ 3000 MHz
Maximum Output Power			6		dBm	0 dB attenuation; 3 × bandwidth/2 offset
Third Order Output Intermodulation Intercept Point	OIP3		27		dBm	600 MHz < f ≤ 3000 MHz
Error Vector Magnitude (3GPP Test Signals)	EVM					
1900 MHz LO			0.6		%	50 kHz RF PLL loop bandwidth
3800 MHz LO			0.53		%	300 kHz RF PLL loop bandwidth

Parameter	Symbol	Min	Typ	Max	Unit	Test Condition/Comment
Observation Receivers	ORx					
Center Frequency		450		6000	MHz	
Gain Range			30		dB	IIP3 improves dB for dB for the first 18 dB of gain attenuation; QEC performance optimized for 0 dB to 6 dB of attenuation only.
Analog Gain Step			0.5		dB	For attenuation steps from 0 dB to 6 dB
Receiver Bandwidth				450	MHz	
Maximum Usable Input Level	P_{HIGH}		-11		dBm	0 dB attenuation; increases dB for dB with attenuation.
Integrated Noise			-58.7		dBFS	450 MHz Integration bandwidth
			-57.5		dBFS	491.52 MHz Integration bandwidth
Third Order Input Intermodulation Intercept Point	IIP3					
Narrow Band						Maximum observation receiver gain; test condition: PHIGH-11 dB/tone
1900 MHz			15		dBm	
2600 MHz			16.5		dBm	
3800 MHz			18		dBm	
Wide Band						IM3 products>130 MHz at baseband; test condition: P_{HIGH} - 11 dB/tone; 491.52 MSPS
1900 MHz			13		dBm	
2600 MHz			11		dBm	
3800 MHz			13		dBm	
Third Order Intermodulation Product	IM3		-70		dBc	600 MHz < f ≤ 3000 MHz
Spurious Free Dynamic Range	SFDR		64		dB	Non IMx related spurs, does not include HDx; (P_{HIGH} − 11) dB input signal
Harmonic Distortion						(P_{HIGH} − 11) dB input signal
Second Order Harmonic Distortion Product	HD2		-80		dBc	In band HD falls within ±100 MHz
			-73		dBc	Out of band HD falls within ±225 MHz
Third Order Harmonic Distortion Product	HD3		-70		dBc	In band HD falls within ±100 MHz
			-65		dBc	Out of band HD falls within ±225 MHz

Parameter	Symbol	Min	Typ	Max	Unit	Test Condition/Comment
Receivers						
Center Frequency				6000	MHz	
Analog Gain Step			30		dB	Attenuator steps from 0 dB to 6 dB
			1		dB	Attenuator steps from 6 dB to 30 dB
Receiver bandwidth				200	MHz	
Maximum Usable Input Level	P_{HIGH}		-11		dBm	0 dB attenuation; increase dB for dB with attenuation; continuous wave = 1800 MHz; corresponds to -1 dBFS at ADC 75 MHz <f≤3000 MHz
Noise Figure	NF		12		dB	0 dB attenuation at receiver port 600 MHz <f≤3000 MHz
Input Third Order Intercept Point	IIP3					
Difference Product	IIP3, d					Two (P_{HIGH} − 9) dB tones near band edge
2600 MHz (Wideband)			17		dBm	
2600 MHz (Midband)			21		dBm	
Sum Product	IIP3, s					Two (P_{HIGH} − 9) dB tones, at bandwidth/6 offset from the LO
2600 MHz (Wideband)			20		dBm	
HD3	HD3		-66		dBc	(P_{HIGH} − 6) dB continuous wave tone at bandwidth/6 offset from the LO 600 MHz < f ≤ 4800 MHz

2.3.2.3.1.2 Hardware Design

For the O-RU the sampling function and frequency conversion function can be performed by the transceiver. The transceiver integrates the ADC, DAC, LO, down converter, up converter, and related functions. The block diagram of transceiver design is shown in Figure 2.3.2-6.

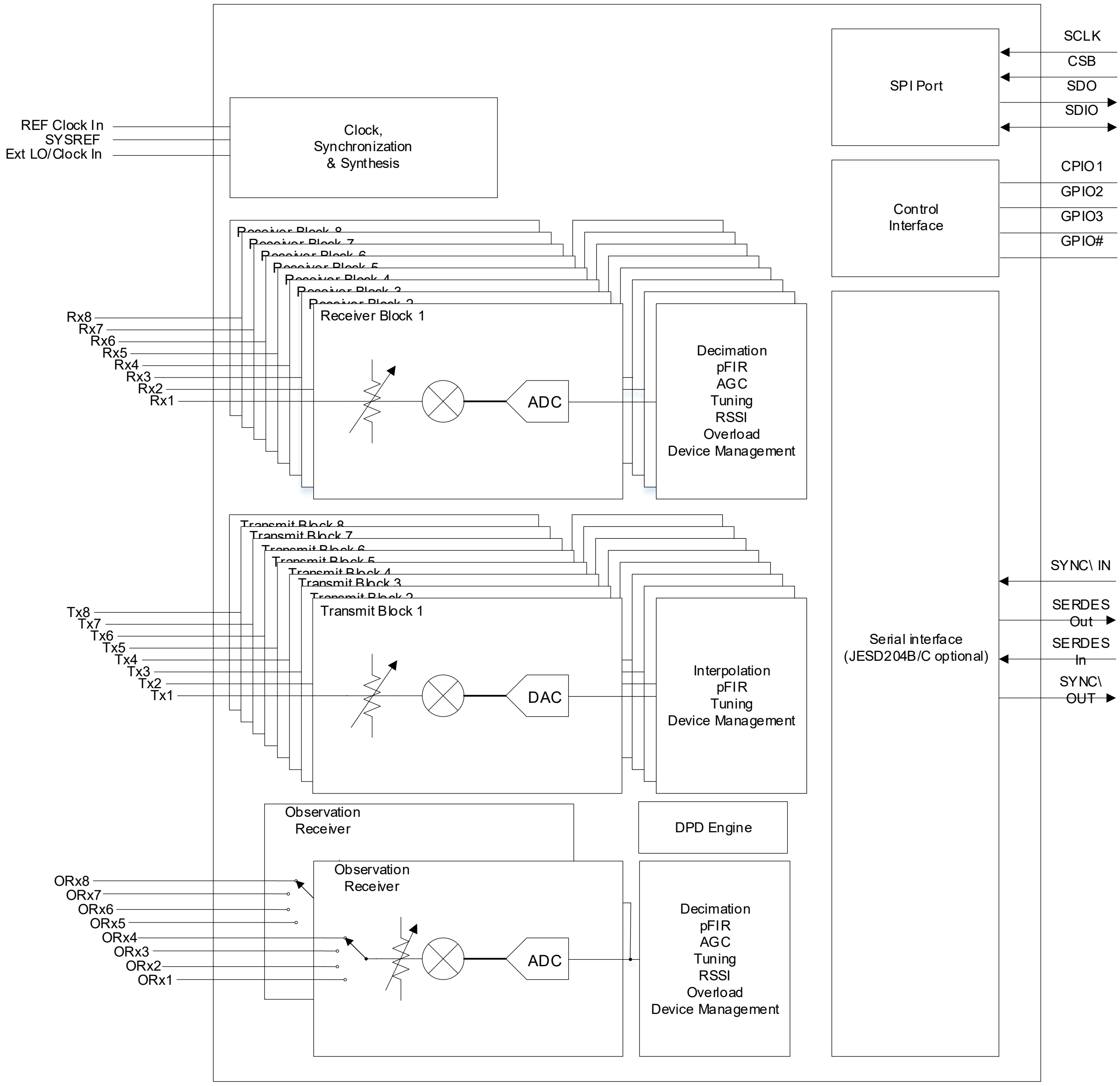


Figure 2.3.2-6 Transceiver with Integrated CFR and DPD

2.3.2.4 Interface Requirements

Table 2.3.2-6 shows the general interfaces required for the O-RAN Processing Unit and Digital Processing Unit based on Figure 2.3.2-4 and Figure 2.3.2-5.

Table 2.3.2-6 Interface Requirements

Item Name	Description
eCPRI Port	High speed bi-directional interface transporting data and control information. Total payload will be determined by the antennas/layers served, bandwidth supported, bit precision, subcarrier spacing, etc. It is served by 4 SFP28 connectors. Actual number of SFP28 transceivers used will depend on RU configuration parameters (e.g., BFP compression bit-width, SRS configuration, etc).
SERDES	A number of serial interface lanes between DPU fabric and transceiver depending on transceiver configuration selected. For example, this could be a 16 lanes high speed JESD204B/C, or some other proprietary interface.
Boot Clock	Clock for any uC or fabric functions required immediately at power on
DDR	Interface for external DDR memory
System Clock	System clock
Clk / PPS out	SyncE / PTP terminated on the FPGA / ASIC

2.3.2.4.1 O-RAN Processing Unit and DPU Design

Table 2.3.2-25 shows the general blocks required for the design. The core elements implemented in this design are required to be sized to support the requirements shown in the OMAC-HAR [6] document. Specifically, the DPU should support up to 200 MHz of occupied BW with up to two 100 MHz component carriers. The OPU-DPU should be capable of processing at least 16 layers in the downlink and 8 layers in the uplink over the whole occupied BW for massive MIMO platforms. Non-massive MIMO platforms may support layers up to the number of antenna paths provided by the hardware. In addition to signal processing, the OPU-DPU should support C, U, S and M plane handling. Major processing blocks in the OPU-DPU are shown in Table 2.3.2-25.

Table 2.3.2-7 OPU-DPU Functional Blocks

Item Name	Description	Cat A	Cat B
FFT & CP-	FFT and Cyclic Prefix removal for uplink processing	Required	Required
iFFT & CP+	iFFT and Cyclic Prefix insertion for downlink processing	Required	Required
PRACH Front End	PRACH front end processor	Required	Required
Channel DDC	Component Carrier Digital Down Converter	Optional	Optional
Channel DUC	Component Carrier Digital Up Converter	Optional	Optional
CFR	Crest Factor Reduction	Optional	Optional
DPD	Digital Pre-Distortion	Optional	Optional
SERDES	High speed serial interface (can be JEDEC open interface or can be proprietary)	Required	Required
C/U/M Plane	Control, User, and management Plane processing	Required	Required
1588v2 Stack	IEEE 1588v2 Synchronization Software Stack	Required	Required
eCPRI	evolved Common Public Radio Interface framer/de-framer	Required	Required
I/Q Compression	I/Q compression/decompression for fronthaul lane rate reduction, including Modulation Compression for Downlink in CatB	Required	Required
Beamforming	Create & steer several beams	Optional	Optional
Pre-coding	Pre-coding	Optional	Required
RE Mapping	Resource Element Mapping to Antenna port	Optional	Required

2.3.2.4.2 Power Amplifier (PA) Reference Design

2.3.2.4.2.1 Hardware Specifications

2.3.2.4.2.1.1 Interface

Table 2.3.2-8 Power Amplifier Interface Specifications

Item Name	Description
RF Enable	The enable input should be compatible with 1.8V logic and tolerate 3.3V as required. A logic high enables the PA. A logic low disables the device and places it in a minimum dissipation mode.
RF Output	RF outputs support 50-ohm single ended to properly interface to a directional coupler, isolator, switch, or antenna.
RF Input	RF inputs should support 50-ohm, single ended match to the transceiver output or preamp.

2.3.2.4.2.1.2 Power Specifications

RF output power required is nominally 10 W (40 dBm). For the whole Tx chain including pre-driver amplifier and driver Amplifier this Tx chain efficiency should be more than 33 % since driver and pre-driver work on less Power O/P and are less efficient.

2.3.2.4.2.1.3 RF Specifications

The PA and driver should have enough gain to boost the transceiver output to the rated power level. The output power should be at least 10W (40 dBm) including the loss of the circulator and antenna filter. The ACLR with DPD and CFR should be better than 47dBc according to the related 3GPP test models.

Table 2.3.2-9 Power Amplifier RF Specifications

Item Name	Band26	Band66	Band48
Gain	15 dB	14.5 dB	14 dB
P3dB	50 dBm	49.7 dBm	49.8 dBm
Input Return Loss	< 12 dB	< 12 dB	< 12 dB
Output Return Loss	< 6 dB	< 6 dB	< 6 dB
Efficiency	> 42%	> 42%	> 45%
Band Flatness	< 0.4 dB	< 0.5 dB	< 0.6 dB
ACLR	<-28 dBc	<-28 dBc	<-29 dBc

2.3.2.4.2.2 Hardware Design

RFin is the input of the PA, RFout is the output of the PA. Vdd and Vbias is the power input of the PA. RFenable is the control pin to disable or enable the PA. The input and output should be matched to 50 ohms to optimize return loss as much as possible. Proper decoupling shall be provided to minimize the impact of supply ripple on the RF output. Figure 2.3.2-7 below details an example of a Power Amplifier reference design.

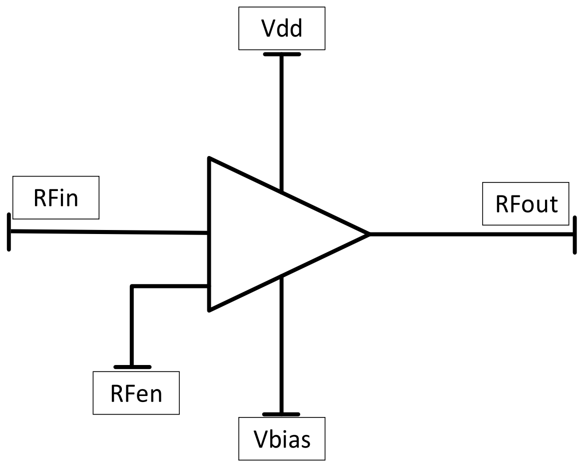


Figure 2.3.2-7 Power Amplifier Reference Design

2.3.2.4.3 Low Noise Amplifier (LNA) Reference Design

2.3.2.4.3.1.1 Interface

Table 2.3.2-10 LNA Interfaces

Item Name	Description
Enable	The enable input should be compatible with 1.8V logic and tolerate 3.3V as required. A logic high enables the LNA. A logic low disables the device and places it in a minimum dissipation mode.
LNA Bypass	The LNA Bypass skips the second stage LNA effectively reducing the overall gain. This would be selected when the input signal compromises the linearity of the amplifier.
RF Out	RF outputs support 50-ohm single ended to properly interface to the RF filtering and Rx input port of transceiver.
TR Switch	The TR switch is used to shunt any reflected RF power that reaches the LNA to a shunt load during the transmit period.
RF Input	RF inputs should support 50-ohm, single ended match to circulator.
Termination	This RF output port shunts any reflected transmit power to an appropriate load during the Transmit period to protect the input of the LNA.

2.3.2.4.3.1.2 Power Specifications

DC power per channel should not exceed 0.5W.

2.3.2.4.3.1.3 RF Performance Specifications

The following table shows the LNA RF specification for O-RU.

Table 2.3.2-11 LNA RF Specifications

Item Name	Band 26	Band 66	Band 48
Gain (High/Low)	45/21 dB	37/17 dB	28/11 dB
NF (High/Low)	0.7/0.7 dB	0.5/0.5 dB	1.1/1.1 dB
OIP3 (High/Low)	35/33 dBm	35/32 dBm	36/30 dBm
OP1dB (High/Low)	21/20 dBm	20/17.5 dBm	19/14dBm
TR Switch Time	na	na	290 nsec

Considering RF output power required is nominally 10W (40 dBm).

2.3.2.4.3.2 Hardware Design

Figure 2.3.2-8 shows the configuration for a dual-stage LNA. The RF inputs and outputs should be properly matched to optimize the noise performance and flatness of the amplifier. The input switch is used to forward any reflected transmit energy to an external load during the transmit period to protect the LNA from damage. The DC supply voltage, VCC, should be properly filtered to prevent power supply noise from entering the LNA and reducing performance.

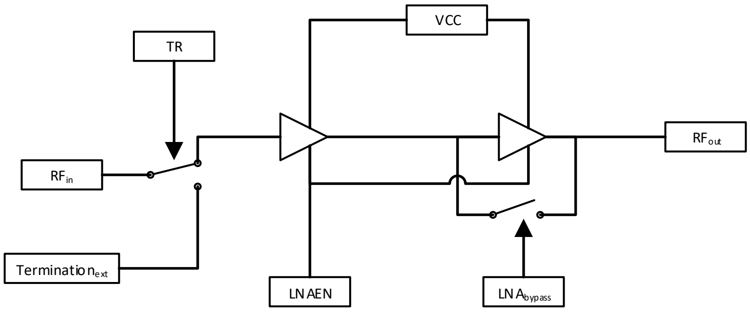


Figure 2.3.2-8 LNA Reference Design

2.3.2.4.4 Circulator Reference Design

For TDD use, the Tx and Rx links work in time duplex. To properly couple the Tx energy to the antenna and the Rx signal to the LNA input, a circulator is used to perform this function given the potentially high RF power being processed through this device. The circulator can be configured as a clockwise or counterclockwise device depending on the layout details.

2.3.2.4.4.1 Hardware Specifications

2.3.2.4.4.1.1 Interface

The circulator includes 3 ports that can function as input or output. If the device is labelled clockwise, then for each input, its nominal output is the next pin rotating right around the device. If it is counterclockwise, the nominal output is the next pin rotating to the left around the device.

2.3.2.4.4.1.2 RF Specifications

The following Table 2.3.2-12 details the Circulator Specifications for O-RU.

Table 2.3.2-12 Circulator Specifications

Item Name	Band 26	Band 66	Band 48
Rotation	Clockwise/Anti Clockwise	Clockwise/Anti Clockwise	Clockwise/Anti Clockwise
Average Power	15 W	15 W	15 W
Peak Power	90 W	90 W	90 W
Impedance	50 Ω	50 Ω	50 Ω
Insertion Loss	≤0.3 dB	≤0.3 dB	≤0.4 dB
Isolation	>20dB	>20dB	>20dB
Return Loss	>20dB	>20dB	>20dB

2.3.2.4.4.2 Hardware Design

The example on the left in Figure 2.3.2-9 is a clockwise oriented device. The RF input enters pin 1 and exits pin 2. Pin 2 can also be used as an input and its output is pin 3. Finally pin 3 can also function as an input with pin 1 being the output. The device on the right is counterclockwise oriented and the signal flow is in the opposite direction.

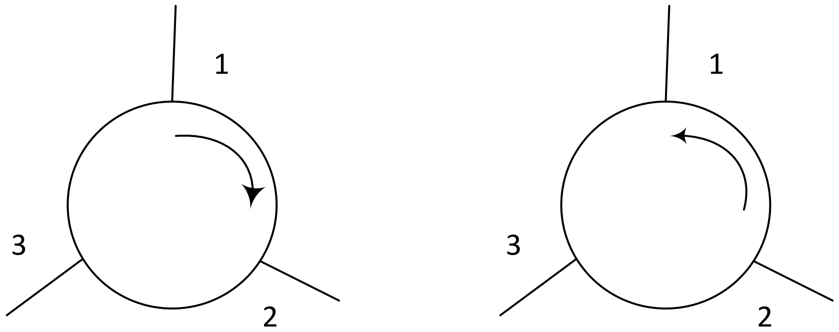


Figure 2.3.2-9 Circulator Reference Design

2.3.2.4.5 Antenna / Phased Array Reference Design

The antenna is a transducer, and it is used to convert electrical signal to the Electromagnetic waves during the transmit mode and vice versa in the reception mode. Based on the radiation pattern characteristics of the antenna, it is broadly classified as omni-directional and directional antennas and examples of such antennas are listed below for reference.

2.3.2.4.5.1 Hardware Specifications for Omnidirectional Antenna

The Table 2.3.2-13 below details the typical electrical omnidirectional antenna specifications for the O-RU.

Table 2.3.2-13 Omnidirectional Antenna Specifications

Frequency band	n26	n66/n70	n48
Antenna Type	Quasi-Omni	Quasi-Omni	Quasi-Omni
VSWR (max)	1.5:1	1.5:1	1.5:1
Power capacity	75W per port, max	75W per port, max	35W per port, max
Gain (avg)	5.0 dBi	7.5 dBi	6.0 dBi
Beam width (vertical)	35º	27º	32º
Polarization	±45º	±45º	±45º
3rd Order PIM, @2x43 dBm	≤-150 dBc	≤-150 dBc	≤-150 dBc
Isolation (cross polarization)	≥25 dB	≥25 dB	≥25 dB

2.3.2.4.5.2 Hardware Design of Omnidirectional Antenna

One possible choice of the omnidirectional antenna is a cannister type with multiple ports including the 4 ports for the 4T4R radio. Conceptual design is shown in the following figure.

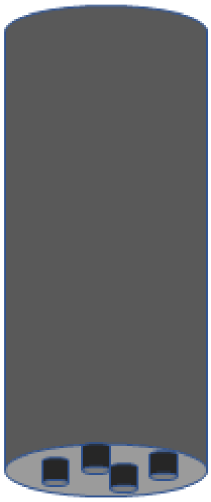


Figure 2.3.2-10 Canister Style Omni Antenna

2.3.2.4.5.3 Hardware Specifications for Directional Antenna

The following table shows the typical electrical directional antenna specifications for the O-RU.

Table 2.3.2-14 Directional Antenna Specifications

Frequency band	n71	n26	n66/n70
Antenna Type	Directional 4T4R	Directional 4T4R	Directional 4T4R
VSWR (max)	1.5:1	1.5:1	1.5:1
Power capacity	250W per port, max	250W per port, max	200W per port, max
Gain (avg)	13.8 dBi	14.5 dBi	18.0 dBi
Beam width (horizontal)	65º	65º	65º
Beam width (vertical)	14º	12º	5º
Polarization	±45º	±45º	±45º
Variable Down Tilt	2-14 º	2-14 º	2-12 º
3rd Order PIM, @2x43 dBm	≤-150 dBc	≤-150 dBc	≤-150 dBc
Isolation (cross polarization)	≥25 dB	≥25 dB	≥25 dB

2.3.2.4.5.4 Hardware Design of Directional Antenna

One possible choice of directional antennas are the radios mounted on the back of the antenna with 4 ports for the 4T4R radio on the back of the antenna just below each radio. Conceptual design is shown in the following figure.

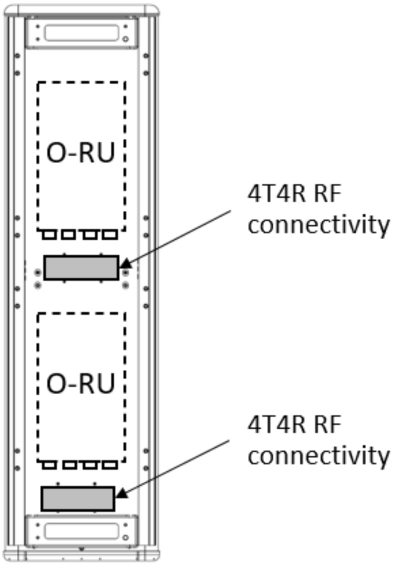


Figure 2.3.2-11 Mounting Arrangement Scenario 1 of 4T4R Radios back of the Directional Antenna

Another possible choice of directional antennas are the radios located away from the antenna with ports for each of the 4T4R radios on the bottom cap of the antenna. Conceptual design is shown in the following figure.

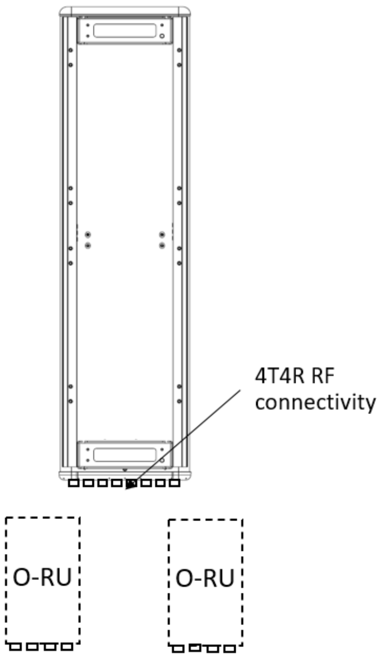


Figure 2.3.2-12 Mounting Arrangement Scenario 2 of 4T4R Radios back of the Directional Antenna

2.3.2.5 Internal Interfaces

The following sections describe potentially useful optional interfaces within the white box. These optional interfaces are not required and may be used at the discrepancy of the manufacturer. This section is provided strictly as information only to better clarify the functionality of these interfaces.

2.3.2.5.1 Optional JESD204 Interfacing

O-RU solutions requiring chip to chip interfacing of high-speed time domain data may benefit from utilization of JEDEC defined standard JESD204 [12]. This interface is specified for chip-to-chip interfacing that provides fast, low latency and time aligned data. The benefit of such an interface is to enable an open standard that allows for devices from multiple vendors to seamlessly interface. From a logistical perspective JESD204 allows multiple devices to communicate over the same link in a multiplex fashion while preserving time alignment and providing deterministic latency. This makes JESD204 a good candidate interface for carrying the signals for multiple antennas, multiple component carriers including I and Q data. This allows for relative simplicity requiring only a few differential pairs along with a 'device clock' and 'sysref' signals to carry and synchronize all time aligned signals.

Figure 2.3.2-13 provides an example of a typical configuration for a JESD204 configuration. The lines labelled 'JESD for UL x2' and 'JESD for DL x2' are the data bearing connections and consist of differential pairs. In this example, there are 2 differential pairs for the UL data (receiver path) and 2 differential pairs for the DL data (transmit path). These connections provide data using an embedded clock and synchronization with a transport layer coding used to ensure sufficient transitions that the embedded clock can be recovered on the receiving end.

The second signal of is labelled 'Device Clock'. The device clock provides the frequency reference to devices in the configuration. Within the scope of JESD204, this signal provides the reference for the interface. In practice, the Device Clock also provide the frequency reference to the individual devices and is usually a multiple of 30.72 MHz. JESD204 connected devices would use this reference for both the interface as well as other frequency services within the device such as sample clocks and other PLLs.

The final signal is the 'Sysref' which enables deterministic latency in the system. Each interconnected device uses this signal in conjunction with 'Device Clock' to determine exactly when the sample edge should occur. There are different ways to configure and use 'Sysref' to achieve different goal. For more details on configuration, see [12].

JESD204 is an evolving standard. Although it has been around for a while, as process technology evolves and data needs grow, JESD has evolved to keep pace with throughput needs. The initial variation supported up to 3.125 Gbps. JESD204A brough speeds up to 6.25 Gbps. JESD204B brought throughput to 12.5 Gbps. Most recently JESD204C offers throughputs up to 32.45 Gbps. Each variety offers at least one option for transport layer coding like 8B/10B as well as a range of other features including sub-channel coding which allows control channels to be passed along with the primary data. Currently most devices support JESD204B and JESD204C. As of this writing, JEDEC is underway defining the next phase of JESD204x which will further increase data rates as demanded by increasing bandwidth in wireless and other application.

Table 2.3.2-15 JESD204 Abridged Lexicon

Parameter	Description
M	Number of converters/converter device (link)
NP (or N')	Total number of bits per sample
S	Samples transmitted/single converter/frame cycle
F	Octets/Frame
K	Number of frames per multi-frame
SC	Sample Clock Rate
L	Number of lanes/converter device (link)

To better comprehend the JESD204 standard, understanding the lexicon is important. There are many parameters that can be used, but those in Table 2.3.2-15 provide the core definitions. Most of these are basic parameters related to common data converter characteristics and include the number of converters, number of bits and sample rate. The remaining parameters are related to the transport layer and physical implementation of the interface.

Combined, the parameters in this lexicon help define the data throughput, number of lanes required as well as the framing of data onto the interfaces. Starting with the payload, it is determined by computing the serial data rate and adding the transport layer (i.e. 8B/10B, 64B/66B etc) to that as defined in the following equation:

$$M \times NP \times SC \times Transport\ Layer\ Coding = Payload\ Rate$$

This equation provides the total payload to be delivered between devices. This data is delivered over one or more lanes. If the payload exceeds the capacity of one lane, additional lanes are added to ensure that the throughput per lane matches the capability of the standard. As highlighted in preceding paragraphs, currently JESD204B supports up to 12.5 Gbps and JESD204C supports up to 32.45 Gbps but future evolutions will include higher lane capacity. Where the Payload Rate exceeds an individual lanes capacity, the total Payload Rate is divided by the lane capacity, currently 12.5 Gbps or 32.45 Gbps, to determine the approximate number of lanes required. The actual number is the next higher integer to ensure that the lane rate is less than the maximum specified or desired. As the standard continues to evolve, this same math will be used to determine the payload and number of lanes, L.

$$L = Celing\left(\frac{Payload\ Rate}{Max\ Lane\ Rate}\right)$$

The final important term is F, the number of octets per frame. In JESD204, an octet represents an encoded 8-bit word. Fundamentally, F defines the frame structure given the number of lanes and the specific payload to deliver. F is defined by the following equation and represents the number of octets per frame:

$$F = \frac{M \times X \times NP}{8 \times L}$$

Figure 2.3.2-13 provides an example 4T4R configuration supporting a typical 100 MHz waveform. The low PHY device processes complex (I & Q data) for each data path and thus there are a total of 8 data streams that must be moved between the digital chip and the data conversion. The bit precision of this data is sufficient to meet the noise requirements of the solution and could be 12 or more bits. For this example, 16-bits will be assumed. To support a 100 MHz bandwidth, a complex data rate of 122.88 MSPS will be used as indicated by Table 2.3.2-16 of [3].

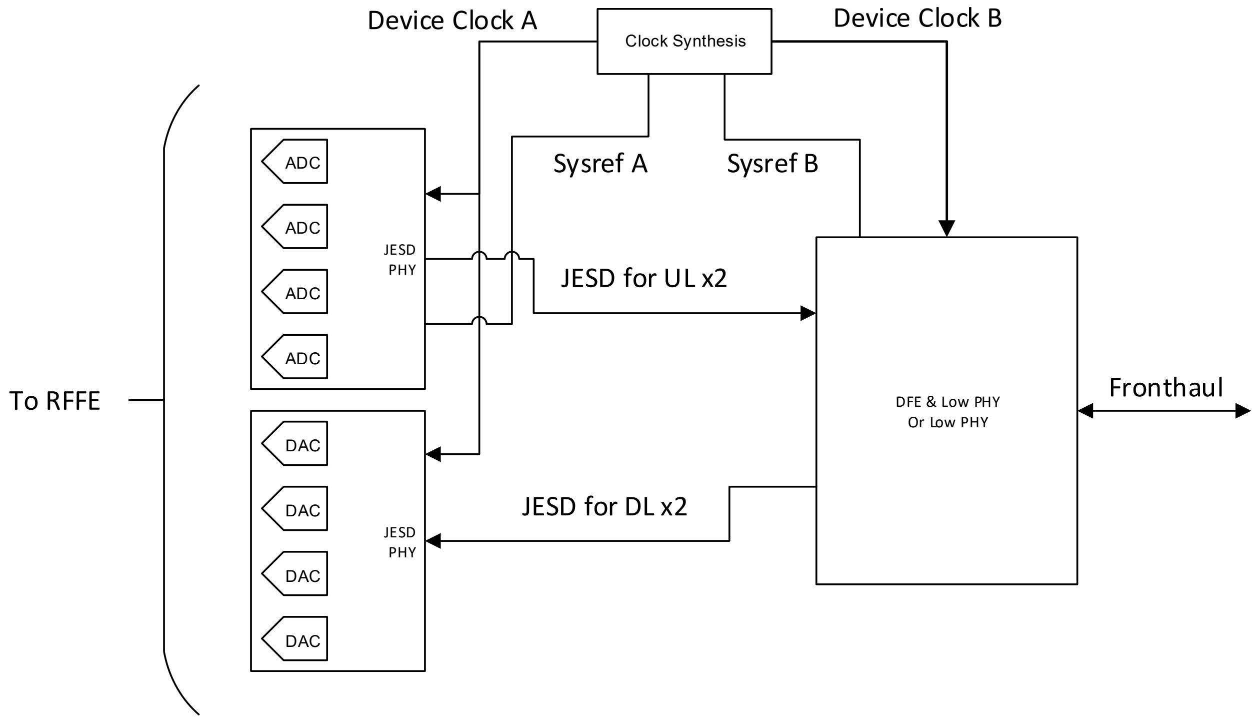


Figure 2.3.2-13 Example 4T4R JESD204 Configuration

Using this information and the equations above and applying 8B10B transport layer coding, the total payload, number of lanes and octets per frame can be computed. Since this is a 4T4R application and each path is represented by a complex stream, 'M' equals 8. 'NP' is 16 bits. 'SC' equals 122.88 MSPS. Total data payload can be computed using the following equation:

$$Payload = 8 \times 16 \times 122.88 \ \times 10^6 \times \frac{10}{8} = 19.6608 Gbps$$

Assuming the interface is based on JESD204B, then the maximum data rate for a single lane would be 12.5 Gbps. Therefore, we need to divide the data into two physical lanes of 9.8304 Gbps each. Hence in this example, 'L' equals 2.

$$L = Ceiling\left(\frac{19.6608\ Gbps}{12.5\ Gbps}\right) = 2$$

Finally, 'F' the number of Octets/Frame is computed to be 8.

$$F = \frac{8 \times 1 \times 16}{8 \times 2} = 8$$

Using this information, the mapping of data can be determined by using online tools such as [13]. These tools describe how the converter data is mapped onto each lane by octet as shown in Figure 2.3.2-14. In this example on lane 0, the first octet conveyed are the most significant bits of converter 0. This is followed by the LSBs in the second octet. This is followed by converter 1, 2 and 3 on lane 0. In parallel, lane 1 provides the data for converters 5 through 7 with the most significant octet leading. In this figure, M#SO represents the logical converter where # identifies the specific converter.

Table 2.3.2-16 4T4R Example Data Mapping

	4 (L = 2, M = 8, F = 8, S = 1, NP = 16)							
	Frame 0							
Lanes	**Octet 0**	**Octet 1**	**Octet 2**	**Octet 3**	**Octet 4**	**Octet 5**	**Octet 6**	**Octet 7**
0	M0S0[15:8]	M0S0[7:0]	M1S0[15:8]	M1S0[7:0]	M2S0[15:8]	M2S0[7:0]	M3S0[15:8]	M3S0[7:0]
1	M4S0[15:8]	M4S0[7:0]	M5S0[15:8]	M5S0[7:0]	M6S0[15:8]	M6S0[7:0]	M7S0[15:8]	M7S0[7:0]

More complicated examples are also supported as shown in Figure 2.3.2-14 where a 64T64R massive MIMO application is described. This configuration supports a 200 MHz iBW and thus requires a complex data rate of 245.76 MSPS. In this case, there are 64 transmit and 64 receive paths. Each of these requires a complex path to the low PHY / beam former device. Therefore 'M' equals 128. Each of these are represented by 16 bits. In this example, JESD204C is assumed and thus 64B/66B coding is used. Thus, the total throughput becomes:

$$Payload = 128 \times 16 \times 245.76 \ \times 10^6 \times \frac{66}{64} = 519.04512 Gbps$$

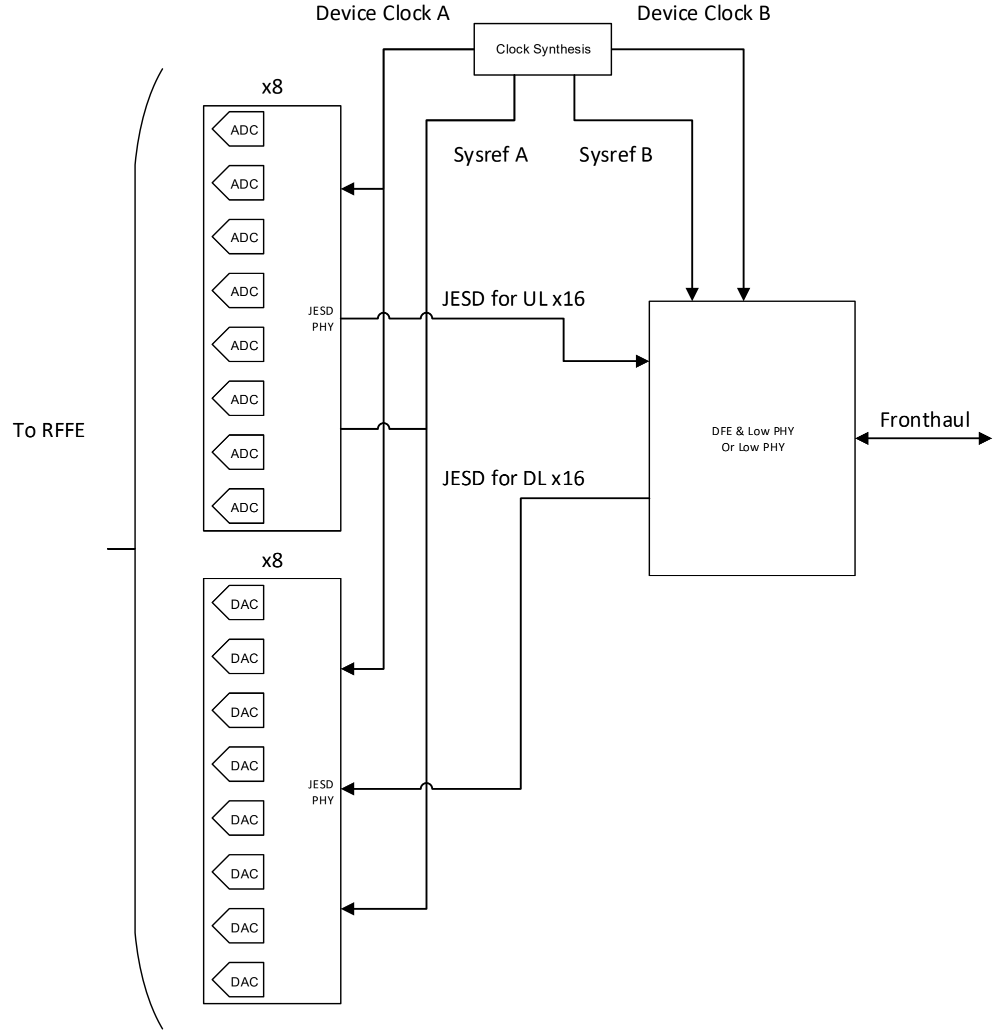


Figure 2.3.2-14 Example 64T64R JESD204 Configuration

JESD204C supports up to 32.45 Gbps and therefore 16 lanes will be required as shown below. In this example, each transceiver device supports 8 channels. Therefore, each radio requires 2 UL and 2 DL JESD lanes which represents a total of 16 lanes in each direction as computed.

$$L = Ceiling\left(\frac{519.04512Gbps}{32.45\ Gbps}\right) = 16$$

Using the equation for 'F' determines the octets per frame to be 16 in this example. The frame structure as defined by JESD204 is shown in Figure 2.3.2-15. As with the previous example, the frame structure is little endian with lane 0 beginning with converter 0, lane 1 beginning with 8, lane 2 with 16 and so forth as shown. Complete details on JESD204 can be found [12].

$$F = \frac{128 \times 1 \times 16}{8 \times 16} = 16$$

	64 (L = 16, M = 128, F = 16, S = 1, NP = 16)															
	Frame 0															
Lanes	Octet 0	Octet 1	Octet 2	Octet 3	Octet 4	Octet 5	Octet 6	Octet 7	Octet 8	Octet 9	Octet 10	Octet 11	Octet 12	Octet 13	Octet 14	Octet 15
0	M0S0[15:8]	M0S0[7:0]	M1S0[15:8]	M1S0[7:0]	M2S0[15:8]	M2S0[7:0]	M3S0[15:8]	M3S0[7:0]	M4S0[15:8]	M4S0[7:0]	M5S0[15:8]	M5S0[7:0]	M6S0[15:8]	M6S0[7:0]	M7S0[15:8]	M7S0[7:0]
1	M8S0[15:8]	M8S0[7:0]	M9S0[15:8]	M9S0[7:0]	M10S0[15:8]	M10S0[7:0]	M11S0[15:8]	M11S0[7:0]	M12S0[15:8]	M12S0[7:0]	M13S0[15:8]	M13S0[7:0]	M14S0[15:8]	M14S0[7:0]	M15S0[15:8]	M15S0[7:0]
2	M16S0[15:8]	M16S0[7:0]	M17S0[15:8]	M17S0[7:0]	M18S0[15:8]	M18S0[7:0]	M19S0[15:8]	M19S0[7:0]	M20S0[15:8]	M20S0[7:0]	M21S0[15:8]	M21S0[7:0]	M22S0[15:8]	M22S0[7:0]	M23S0[15:8]	M23S0[7:0]
3	M24S0[15:8]	M24S0[7:0]	M25S0[15:8]	M25S0[7:0]	M26S0[15:8]	M26S0[7:0]	M27S0[15:8]	M27S0[7:0]	M28S0[15:8]	M28S0[7:0]	M29S0[15:8]	M29S0[7:0]	M30S0[15:8]	M30S0[7:0]	M31S0[15:8]	M31S0[7:0]
4	M32S0[15:8]	M32S0[7:0]	M33S0[15:8]	M33S0[7:0]	M34S0[15:8]	M34S0[7:0]	M35S0[15:8]	M35S0[7:0]	M36S0[15:8]	M36S0[7:0]	M37S0[15:8]	M37S0[7:0]	M38S0[15:8]	M38S0[7:0]	M39S0[15:8]	M39S0[7:0]
5	M40S0[15:8]	M40S0[7:0]	M41S0[15:8]	M41S0[7:0]	M42S0[15:8]	M42S0[7:0]	M43S0[15:8]	M43S0[7:0]	M44S0[15:8]	M44S0[7:0]	M45S0[15:8]	M45S0[7:0]	M46S0[15:8]	M46S0[7:0]	M47S0[15:8]	M47S0[7:0]
6	M48S0[15:8]	M48S0[7:0]	M49S0[15:8]	M49S0[7:0]	M50S0[15:8]	M50S0[7:0]	M51S0[15:8]	M51S0[7:0]	M52S0[15:8]	M52S0[7:0]	M53S0[15:8]	M53S0[7:0]	M54S0[15:8]	M54S0[7:0]	M55S0[15:8]	M55S0[7:0]
7	M56S0[15:8]	M56S0[7:0]	M57S0[15:8]	M57S0[7:0]	M58S0[15:8]	M58S0[7:0]	M59S0[15:8]	M59S0[7:0]	M60S0[15:8]	M60S0[7:0]	M61S0[15:8]	M61S0[7:0]	M62S0[15:8]	M62S0[7:0]	M63S0[15:8]	M63S0[7:0]
8	M64S0[15:8]	M64S0[7:0]	M65S0[15:8]	M65S0[7:0]	M66S0[15:8]	M66S0[7:0]	M67S0[15:8]	M67S0[7:0]	M68S0[15:8]	M68S0[7:0]	M69S0[15:8]	M69S0[7:0]	M70S0[15:8]	M70S0[7:0]	M71S0[15:8]	M71S0[7:0]
9	M72S0[15:8]	M72S0[7:0]	M73S0[15:8]	M73S0[7:0]	M74S0[15:8]	M74S0[7:0]	M75S0[15:8]	M75S0[7:0]	M76S0[15:8]	M76S0[7:0]	M77S0[15:8]	M77S0[7:0]	M78S0[15:8]	M78S0[7:0]	M79S0[15:8]	M79S0[7:0]
10	M80S0[15:8]	M80S0[7:0]	M81S0[15:8]	M81S0[7:0]	M82S0[15:8]	M82S0[7:0]	M83S0[15:8]	M83S0[7:0]	M84S0[15:8]	M84S0[7:0]	M85S0[15:8]	M85S0[7:0]	M86S0[15:8]	M86S0[7:0]	M87S0[15:8]	M87S0[7:0]
11	M88S0[15:8]	M88S0[7:0]	M89S0[15:8]	M89S0[7:0]	M90S0[15:8]	M90S0[7:0]	M91S0[15:8]	M91S0[7:0]	M92S0[15:8]	M92S0[7:0]	M93S0[15:8]	M93S0[7:0]	M94S0[15:8]	M94S0[7:0]	M95S0[15:8]	M95S0[7:0]
12	M96S0[15:8]	M96S0[7:0]	M97S0[15:8]	M97S0[7:0]	M98S0[15:8]	M98S0[7:0]	M99S0[15:8]	M99S0[7:0]	M100S0[15:8]	M100S0[7:0]	M101S0[15:8]	M101S0[7:0]	M102S0[15:8]	M102S0[7:0]	M103S0[15:8]	M103S0[7:0]
13	M104S0[15:8]	M104S0[7:0]	M105S0[15:8]	M105S0[7:0]	M106S0[15:8]	M106S0[7:0]	M107S0[15:8]	M107S0[7:0]	M108S0[15:8]	M108S0[7:0]	M109S0[15:8]	M109S0[7:0]	M110S0[15:8]	M110S0[7:0]	M111S0[15:8]	M111S0[7:0]
14	M112S0[15:8]	M112S0[7:0]	M113S0[15:8]	M113S0[7:0]	M114S0[15:8]	M114S0[7:0]	M115S0[15:8]	M115S0[7:0]	M116S0[15:8]	M116S0[7:0]	M117S0[15:8]	M117S0[7:0]	M118S0[15:8]	M118S0[7:0]	M119S0[15:8]	M119S0[7:0]
15	M120S0[15:8]	M120S0[7:0]	M121S0[15:8]	M121S0[7:0]	M122S0[15:8]	M122S0[7:0]	M123S0[15:8]	M123S0[7:0]	M124S0[15:8]	M124S0[7:0]	M125S0[15:8]	M125S0[7:0]	M126S0[15:8]	M126S0[7:0]	M127S0[15:8]	M127S0[7:0]

Figure 2.3.2-15 64T64R Example Data Mapping

2.3.3 Synchronization and Timing

The clock and synchronization are determined by the PTP and SyncE signals transported by ethernet L2 via the front haul in conjunction with clock recovery in the hardware. Together these provide the local time of day and frequency reference for the O-RU.

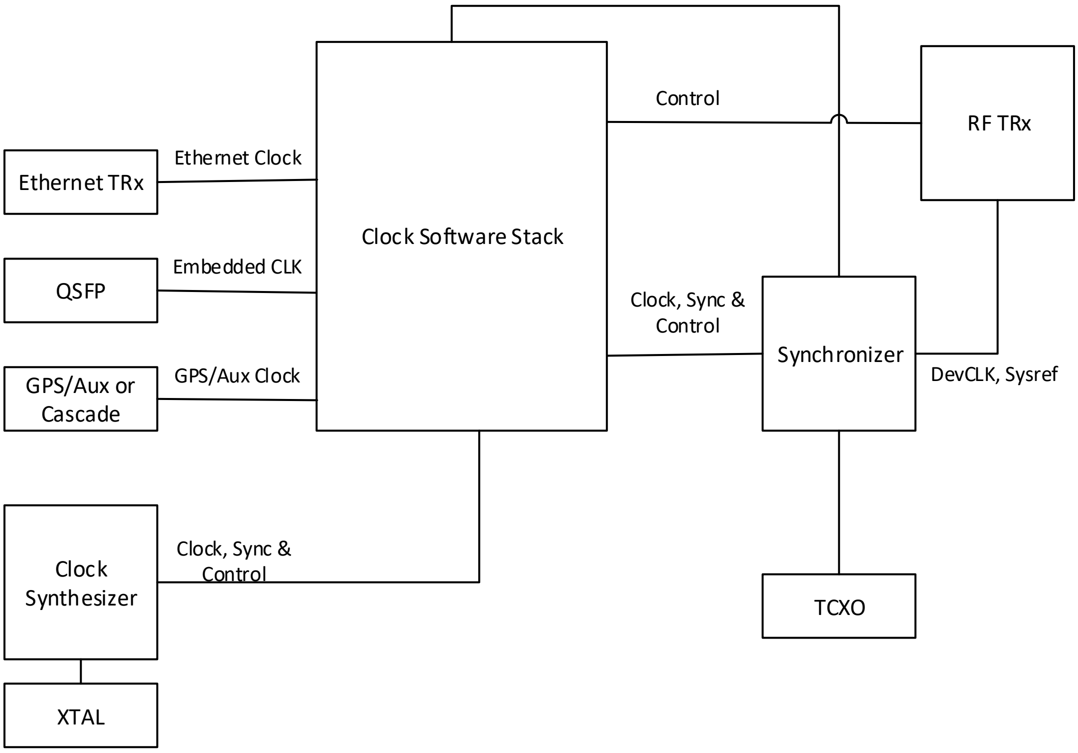


Figure 2.3.3-1 Clock and Synchronization Functional Block Diagram

2.3.3.1 Hardware Specifications

2.3.3.1.1 Interface

Table 2.3.3-1 Synchronization and Timing Interface Specifications

Item Name	Description
Clock Inputs	Depending on the configuration, the following clock inputs may exist: QFSP, Ethernet, GPS, Cascaded unit clocks, Synchronized Clock.
Clock Outputs	Depending on functional requirements, the following clocks may be delivered: QFSP, Ethernet, DDR, CPU, RF Transceiver clock, Synchronizer reference.
Clock Control	SPI or I2C to control the clock synthesizer and synchronizer.
Clock Sync	Logic signal to phase align and time align the synchronizer with master.
Crystal Reference	One or more reference oscillators to provide required phase noise and short/long term stability.

2.3.3.1.2 Performance Specifications

See device synchronizer performance specifications in Table 2.3.3-2

2.3.3.1.3 Synchronizer

The purpose of the Synchronizer is to function as part of the local clock recovery in conjunction with an IEEE1588v2 software stack to provide local time of day and to generate a local synchronous reference signals to be used for synthesis of necessary clocks to other functions within the O-RU.

2.3.3.1.3.1 Hardware Specifications

2.3.3.1.3.1.1 Interface

Table 2.3.3-2 Synchronizer Interface Specifications

Item Name	Description
REFA, REFAA	differential or single ended reference pulse inputs.
REFB, REFBB	differential or single ended reference pulse inputs.
Status and Control	control core operation and report on state of the PLL.
XOA, XOB	complimentary reference inputs or reference crystal connections
Output0 and Output0N & Output1 and Output1N	reference outputs to feed further synthesis, transceiver, or logic blocks.

2.3.3.1.3.1.2 Performance Specifications

Table 2.3.3-3 Synchronizer Performance Specifications

Item Name	Description	Notes
10 Hz Offset	−77 dBc/Hz	Device configuration: fOSC = 52 MHz crystal, fREF = 30.72 MHz, fVCO = 2457.6 MHz, fOUT = 245.76 MHz, DPLL BW = 50 Hz, internal zero delay operation
100 Hz Offset	−93 dBc/Hz	
1 kHz Offset	−114 dBc/Hz	
10 kHz Offset	−125 dBc/Hz	
100 kHz Offset	−130 dBc/Hz	
1 MHz Offset	−140 dBc/Hz	
10 MHz Offset	−156 dBc/Hz	

2.3.3.1.3.2 Hardware Design

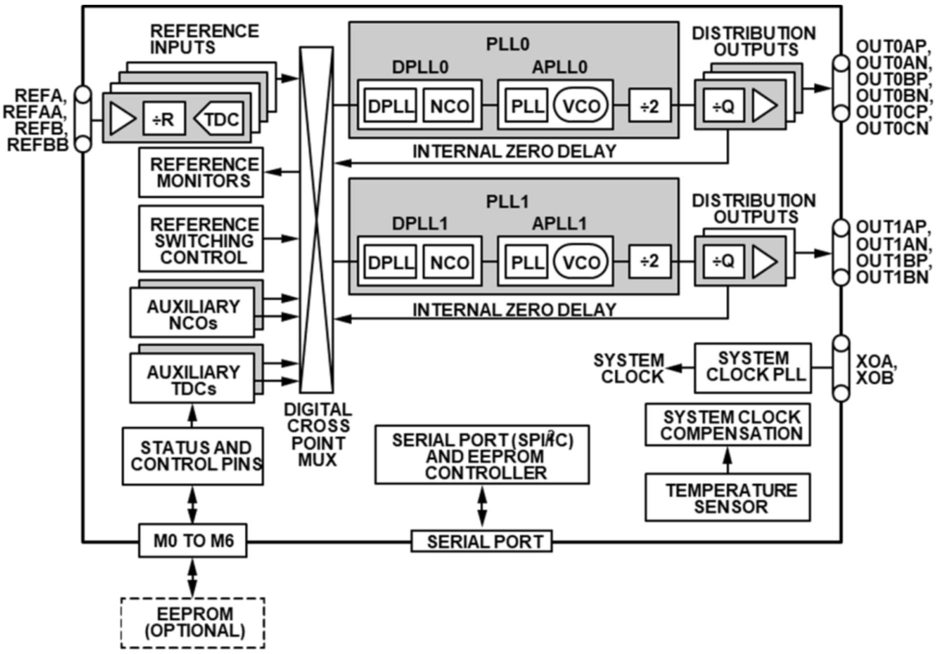


Figure 2.3.3-2 Synchronizer Block Diagram

2.3.3.1.4 Reference Synthesizer

The purpose of the clock synthesizer is to generate low jitter, high stability reference clocks as needed throughout the design.

2.3.3.1.4.1 Hardware Specifications

2.3.3.1.4.1.1 Interface

Table 2.3.3-4 Synthesizer Interface Specifications

Item Name	Description
Reference Clock(s) Input	The clock inputs device should receive a stable reference from a crystal, network reference or other suitable source. More than one input is allowed that may be selected between as required by the system.
Control	Control of the transceiver is by way of 3 or 4 wire SPI or I2C functioning as a slave. Support for 1.8V control is required and tolerance of 3.3V is preferred. Typically, the clock devices will be part of an IEEE1588v2 protocol loop controlled by way of this control interface or other GPIO lines as required by the hardware implementation.
Clock Outputs	One or more clock outputs are required to drive the digital device and transceiver clock inputs. Each output should be independently programmable in frequency to suit the application. The outputs should nominally be differential to preserve clocking edge and to maximize tolerance of high common mode signals.

2.3.3.1.4.1.2 Performance Specifications

Table 2.3.3-5 TCXO Clock Performance Specifications

Item Name	Description
Frequency	25 MHz
Stability	+/- 0.28 ppm
10 Hz Offset	-90 dBc/Hz
100 Hz Offset	−120 dBc/Hz
1 kHz Offset	−140 dBc/Hz
10 kHz Offset	−151 dBc/Hz
100 kHz Offset	−152 dBc/Hz
1 MHz Offset	−154 dBc/Hz

2.3.3.1.4.2 Hardware Design

Figure 2.3.3-3 depicts the hardware design of the Synthesizer Block

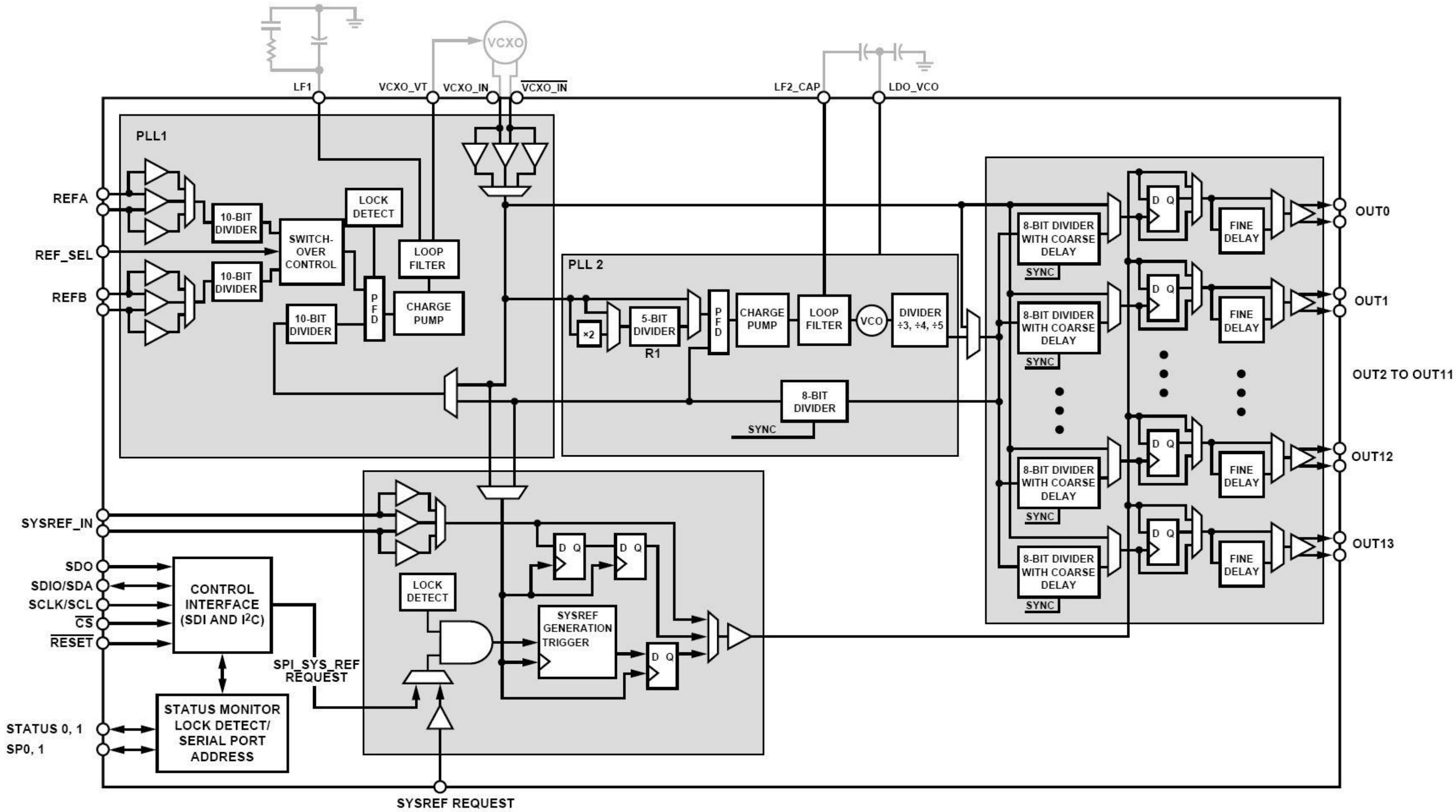


Figure 2.3.3-3 Synthesizer Block Diagram

2.3.4 External Interface Ports

2.3.4.1 Hardware Specifications

The following table shows the external ports or slots provided by the O-RU. External connections can be aggregated into a composite connector to minimize the points of environmental ingress as required for a specific deployment.

Table 2.3.4-1 External Port List

Port Name	Feature Description
Fronthaul interface	One or more 25Gbps optical interfaces including a primary link and optionally a second link for redundancy or daisy chain.
Debug interface	RJ45 for maintenance access.
Power interface	Two pin male panel mount connectors (x1)
RF Interface	4.3-10+ IP67 connector (x4) [7]/M-LOC cluster/multiport RF connector [8]

2.3.4.2 Hardware Design for External Interfaces

2.3.4.2.1 Fronthaul Interface

The fronthaul interface to the OMAC is provided by an optical fiber or fibers connected to the chassis by way of a SFP or SFP+ cage connector. In addition, an IP67 connector seal should be selected to provide appropriate environmental protection.

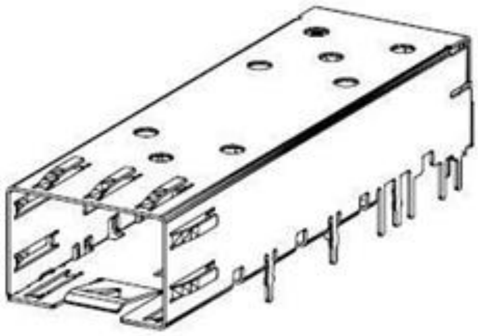


Figure 2.3.4-1 SFP+ case and connector

2.3.4.2.2 Debug Interface

The local maintenance port or debug port is by way of a standard ethernet connector. The RJ45 Ethernet connector are standardized components and available from multiple vendors. The following figure describes the general form factor of one such RJ45 connector.

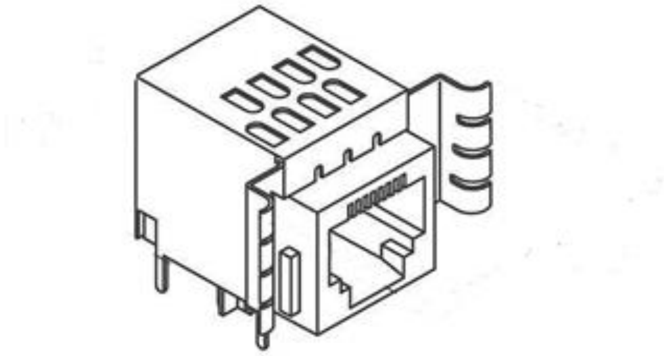


Figure 2.3.4-2 RJ45 Interface

2.3.4.2.3 Power Interface

The standard power interface is defined by a 2-pin panel mount connector as shown in the following figure.

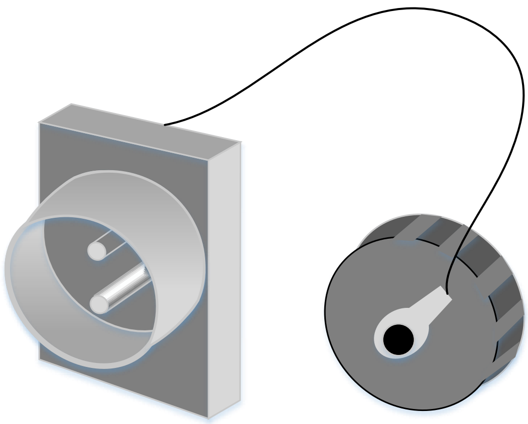
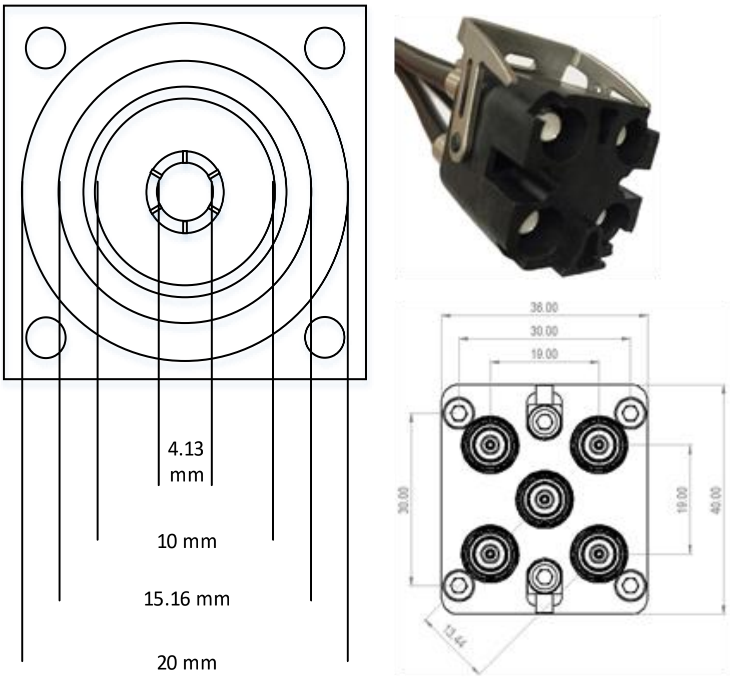


Figure 2.3.4-3 Power Interface

2.3.4.2.4 RF Interface

The standard RF interfaces are the 4.3-10+ Low PIM RF connector and the M-LOC cluster/multiport connector as shown in the following figure.



(a) (b)

Figure 2.3.4-4 (a) 4.3-10+ RF Connector [7] and (b) M-LOC Cluster/multiport connector [8]

2.3.5 Mechanical

One such mechanical structure is shown in the following figure. The shell should provide appropriate environmental protection, RF shielding and passive thermal dissipation. Internally, thermal pads are used to contact local heat sources and transfer that heat to the frame which in turn dissipates the heat through bulk metal and fins used to increase the effective surface area. Other enclosures are possible.

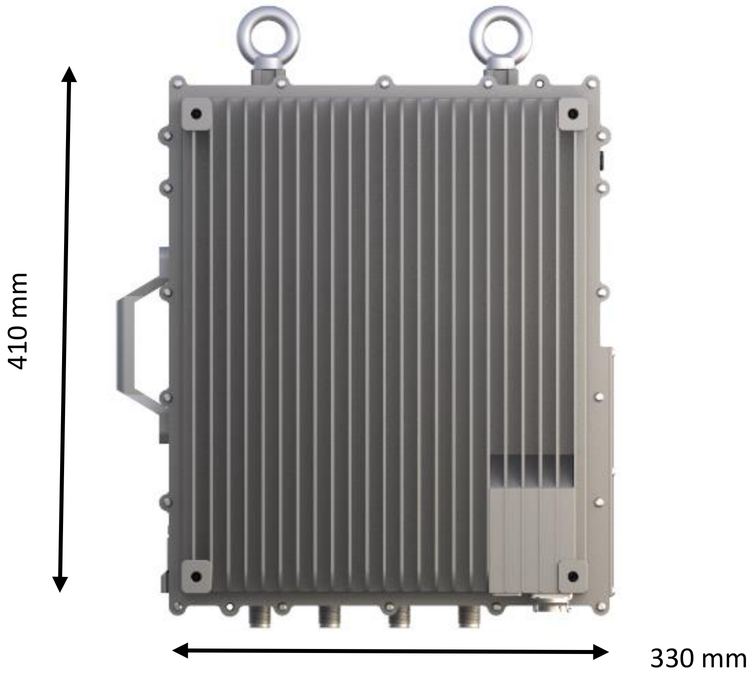


Figure 2.3.5-1 Mechanical Enclosure Front View

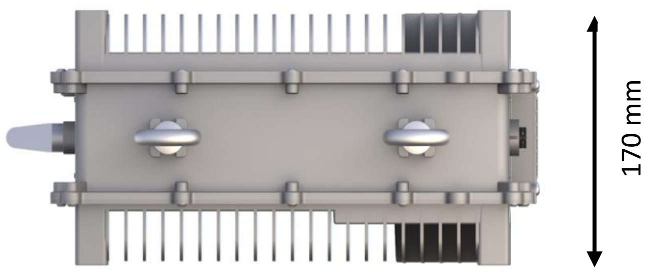


Figure 2.3.5-2 Mechanical Enclosure Top View

2.3.6 Power Unit

For the Outdoor Macrocell, the power solution of the O-RU is provided externally from a telco -48V supply. Internally, the telco supply is converted and distributed to devices using appropriate DC-DC and LDO converters. Optionally the unit can be powered by AC line power which is not shown here.

2.3.6.1 Hardware Specifications

The power unit is responsible for isolating and converting the telco supply to appropriate voltages for distribution to individual supply domains in the O-RU. The worst-case dissipation is outlined below. The RF front end is broken out separately since different deployment scenarios will require different frequencies and power levels resulting in significantly different total dissipation. Total Power is the sum of transceiver module section and RF front end module section.

Table 2.3.6-1 Transceiver Module Specifications

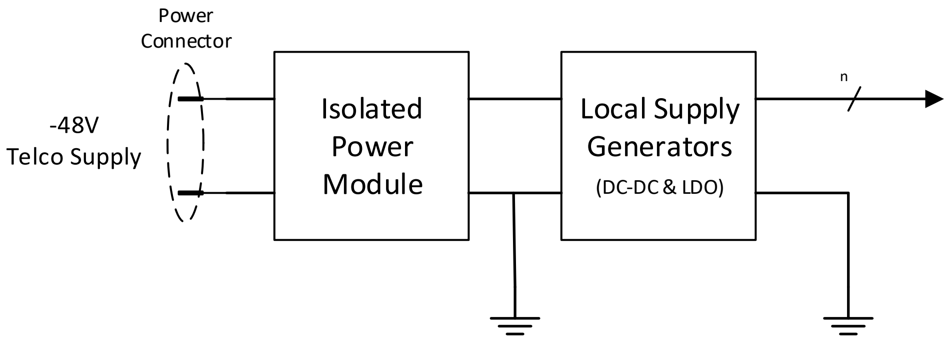
Module	4T4R (Watts)		Notes
FPGA	25 W		
Transceiver	10 W		
Synchronization & Clocking	1.8 W		
DDR	1.8 W		4 modules
Others	6 W		
SFP28 Module	4 W		2 modules
total device consumption	48.6 W		
O-RU Low PHY Transceiver power consumption	54 W		90% power efficiency

Table 2.3.6-2 RF Front End Module Specifications

Module	4T4R (Watts)		Notes
PA	330-350		40W O/P at Antenna Port
PA Drive Amp	30-35		2W O/P Power
LNA	1.5 – 2.0		For 4 chains
Others	10-13		Gain Block and Switch
total device consumption	344.5 – 400		80% Duty Cycle
O-RU RF Front End power consumption	495		Total Power Consumption

2.3.6.2 Hardware Design

The block diagram of the power unit is shown in Figure 2.3.6-1.



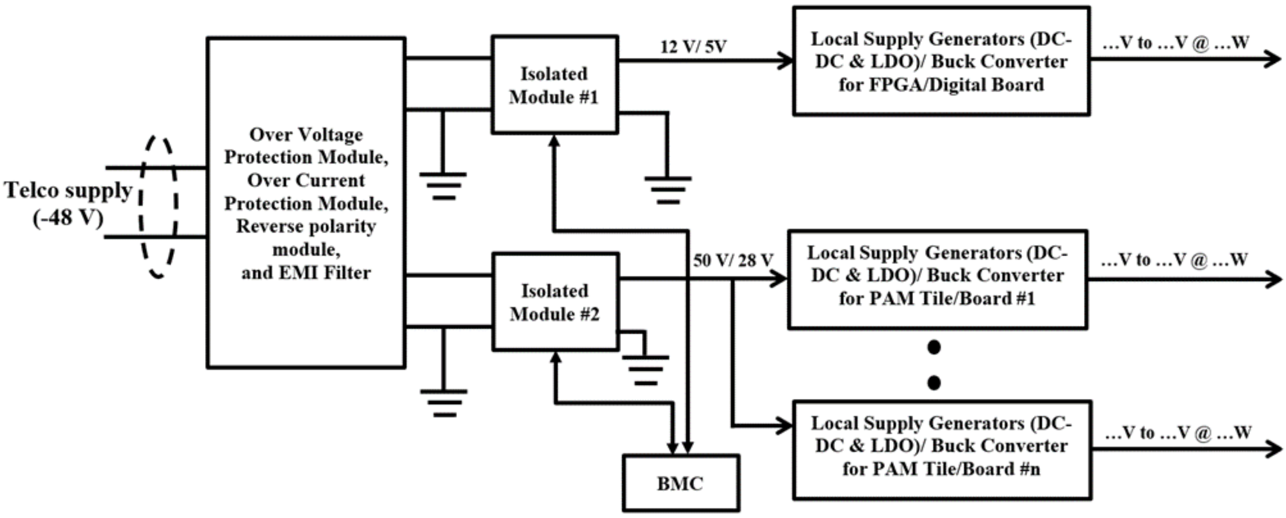


Figure 2.3.6-1 Power Reference Design

2.3.7 Thermal

Heat dissipation in the O-RU is widely done through passive cooling via three modes: conduction, convection, and radiation. These mechanisms are accomplished with the help of heat sink, heat pipes and vapor chamber etc.

The transfer of heat though conduction can usually be described by Fourier's law, which is stated

$$q=(k/s)A.dT$$

k is the thermal conductivity of the material, and has units of W/m·K, s = material thickness, A = heat transfer area, dT = temperature difference between inside and outside wall.

Table 2.3.7-1 Thermal Conductivity of common Metals [9]

Type of Material	Thermal Conductivity (W/m,K)
Diamond	2300
Pure Al	237
Hard Aluminum 4.5% Cu	315-317
Au	411-429
Iron	80
Tin	67

The heat sinks transfer heat to space by thermal convection and radiation. For thermal convection, it can be calculated by the Newton's Equation:

$$Q_d = \alpha_c A \Delta t = \alpha_c A(t_\omega - t_f)$$

where,
Q_d is the convention heat (W), t_ω is the surface temperature of object (K), t_f is the temperature of airflow (K), A is the surface area of the object involved in heat transfer (m2), Δt is the temperature difference between the surface temperature of object and airflow (K), α_c is convective heat transfer coefficient (W/m^2.K).

Table 2.3.7-2 Convective heat transfer coefficient [10]

Flow Type	Heat transfer coefficients (W/m^2. K)
Forced convection; low speed flow of air over a surface	10
Forced convection; moderate speed flow of air over a surface	100
Forced convection; moderate speed crossflow of air over a cylinder	200
Forced convection; moderate flow of water in pipe	3000

For thermal radiation, it can be calculated by the following Equation:

$$Q_r = \varepsilon \delta A(t_\omega^4 - t_f^4)$$

where, Q_r is the radiation heat (W), ε is the thermal emissivity, δ is the Stefan– Boltzmann constant, its value is 5.67032 x10 − 8 W/ (m − 2. K − 4). From these Equations, we can see that the increase of surface area A and thermal emissivity ε will improve the heat dissipation.

Different materials can have widely different emissivity values within the range of 0 to 1.00.

Table 2.3.7-3 Emissivity value of different materials [11]

Material	Emissivity Value (ε)
Aluminum: anodized	0.77
Aluminum: polished	0.05
Brass: highly polished	0.03
Brass: oxidized	0.61
Carbon: candle soot	0.95
Carbon: graphite, filed surface	0.98
Copper: polished	0.05
Copper: oxidized	0.65

2.3.8 Environmental and Regulations

The O-RU hardware system is RoHS Compliant. The unit is passively cooled.

Table 2.3.8-1 Environmental Features

Item Name	Description	Notes
Operating Temperature	-40°C to +55°C	Cold and hot start
Operating Relative Humidity	5% to 95%	
Atmospheric Pressure	70 to 106Kpa	

2.4 Integrated O-DU & O-RU (gNB-DU) Reference Design

The integrated hardware consists of O-DU & O-RU in a single box. The integrated O-DU & O-RU is the white box hardware platform that performs the O- DU functions such as upper L1 and lower L2 functions along with O-RU. The integrated small cell (gNB-DU) shall support split option – 2 (F1 Interface) towards gNB-CU and supports FAPI standard logical interface between Layer2 and Layer1 running on different HW options.

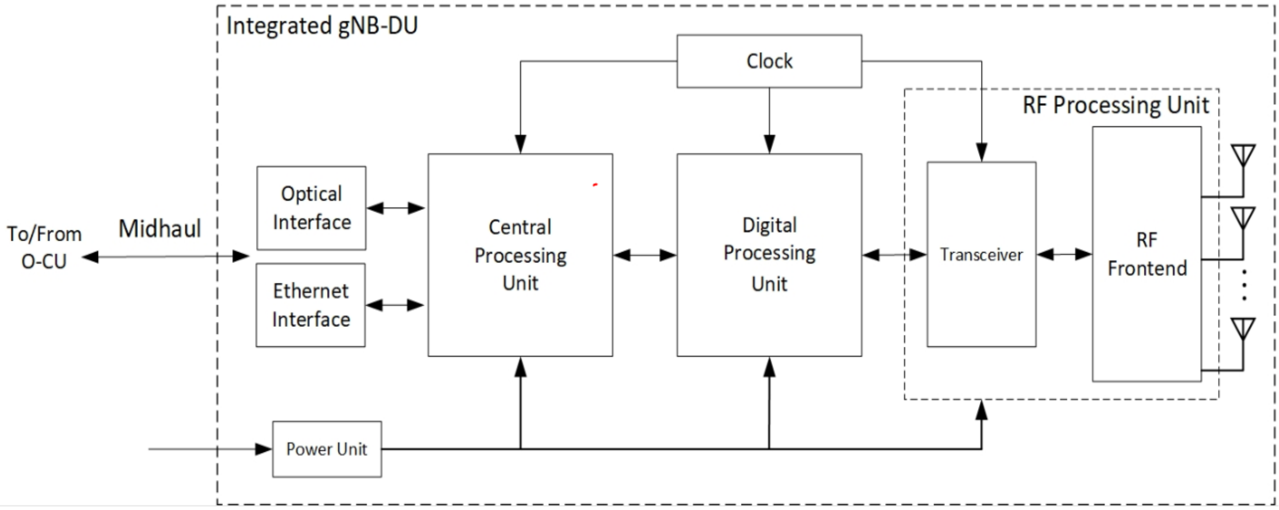


Figure 2.3.8-1 Integrated O-DU & O-RU (gNB-DU) Reference Design

2.4.1 O-DU Portion of Integrated Reference Design

In this Integrated architecture, several functions are included such as DDR4, QSPI, 1G Ethernet, SFP, PCIe, JTAG, UART, and EMMC.

2.4.1.1 O-DU High-Level Functional Block Diagram

In the following O-DU portion in the integrated architecture it shows the interconnection of the Components and external interfaces. O-DU portion is connected to O-RU portion via PCIe interface.

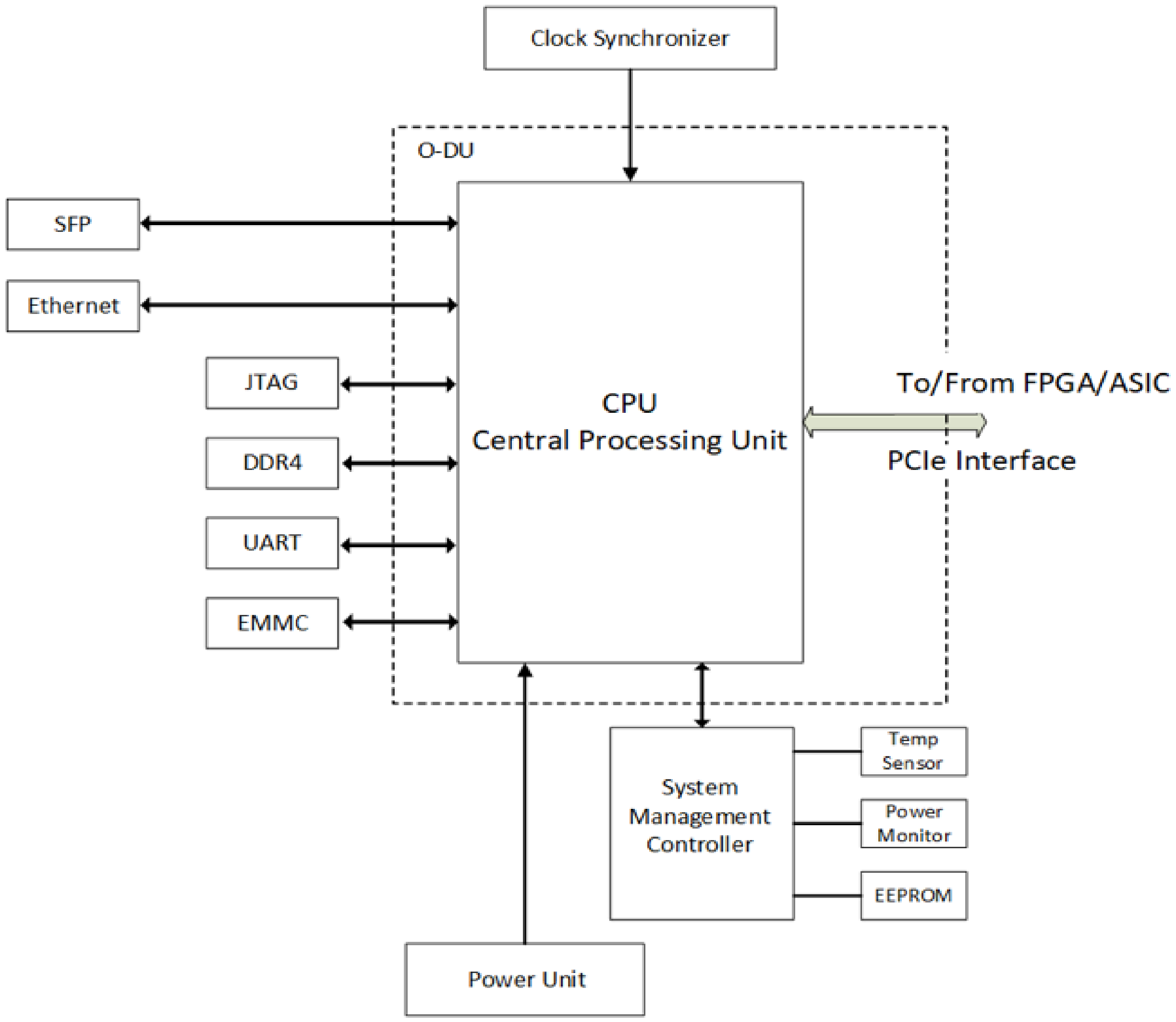


Figure 2.4.1-1 Functional Block Diagram For O-DU portion

2.4.1.2 O-DU Hardware Components

[Intentionally left blank awaiting future contributions]

2.4.1.2.1 Digital Processing Unit

2.4.1.2.2 Hardware Accelerator (if required by design)

2.4.1.2.2.1 Accelerator Design 1

2.4.1.2.2.2 Accelerator Design 2

2.4.1.3 Synchronization and Timing

2.4.2 O-RU Portion of Integrated Reference Design

In this Integrated architecture the O-RU portion contains the Digital Processing Unit Processing Unit, including the Transceiver, RFFE and other RF components, Clock and Synchronization & Power Unit. Digital processing unit is connected to O-DU portion NPU through PCIe interface.

2.4.2.1 O-RU High-Level Functional Block Diagram

This is high level functional block diagram for O-RU portion in the Integrated architecture.

It also highlights the internal/external interfaces that are required.

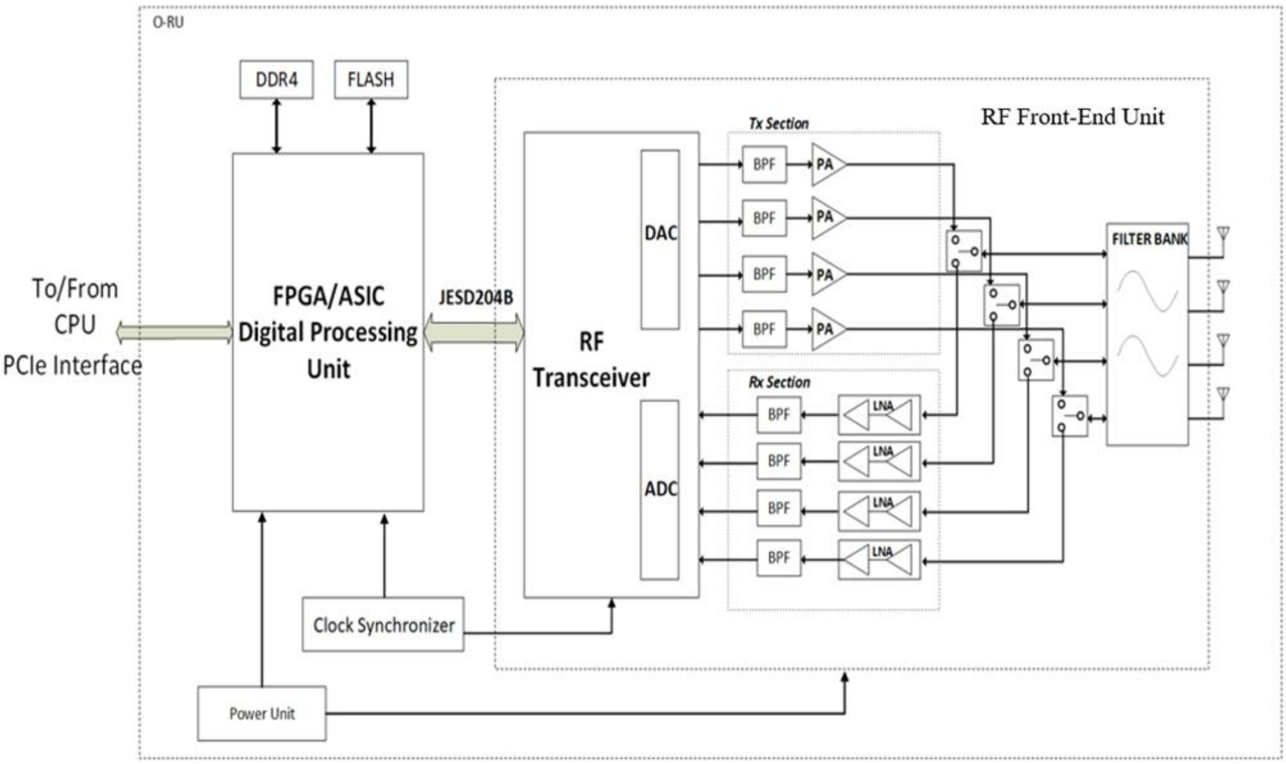


Figure 2.4.2-1 Functional Block Diagram for O-RU portion

2.4.2.2 O-RU Hardware Components

[Intentionally left blank awaiting future contributions]

2.4.2.2.1 Digital Processing Unit

2.4.2.2.2 RF Processing Unit

2.4.2.2.2.1 Transceiver Reference Design

2.4.2.2.2.2 Power Amplifier (PA) Reference Design

2.4.2.2.2.3 Low Noise Amplifier (LNA) Reference Design

2.4.2.2.2.4 RF Switch Reference Design

2.4.2.2.2.5 Antenna / Phased Array Reference Design

Phased array antenna reference design is shown in the Figure 2.4.2-2

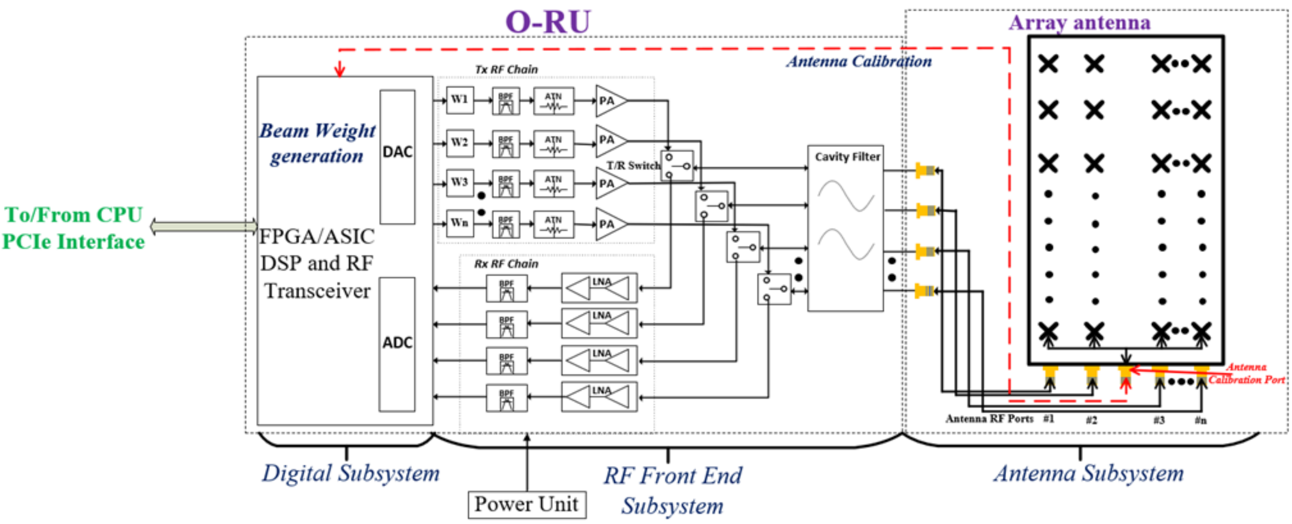


Figure 2.4.2-2 Phased array antenna reference design

It has the following functional blocks:

1. Digital Front-end Block:

It is responsible for processing the baseband data and controlling the beams of the antenna. It has two basic functional blocks that are described below.

   a. FPGA/Baseband processor - To generate/apply the weights (Amplitude and phase) to the phased array antenna elements based on the beam characteristics. Also, it is responsible for scheduling the data/control symbols in each transceiver chain.

   b. DSP/Algorithm block: It works together with the FPGA to generate the weight matrix and map the same to the corresponding antenna element.

2. RF Front-end Block:

It has attenuator, amplifiers (PA in the Tx and LNA in the Rx chain), switches and duplexer (to switch between Tx/Rx chain or Downlink/Uplink)

3. Plurality of antenna elements:

Two or more antenna elements are there in the phased array antenna to attain the below desired beam characteristics upon application of the corresponding beam weights, which are estimated based on the CSI/SRS measurements.

   a. Beam splitting (Hard and Soft) - Twin beam, Quad beam etc...
   b. Beam steering
   c. Plurality of beams (Service beam (Sectoral) and multiple narrow beams)
   d. Beam refinement
   e. Generating null in the desired direction
   f. Tilting of the beams

2.4.2.2.3 Synchronization and Timing

2.4.3 External Interface Ports

2.4.4 O-DU/O-RU Firmware (if required by design)

2.4.5 Mechanical

2.4.6 Power Unit

2.4.7 Thermal

2.4.8 Environmental and Regulations

2.5 O- RU_{7-2x} Massive MIMO Hardware Reference Design

Massive Multiple Input and Multiple Output (mMIMO), as one of the key enablers of 5G NR that enhances the user experience by employing the spatial multiplexing, beamforming, and spatial diversity methods. This $O\text{-}RU_{7-2x}$ consists of several digital baseband and RF processing units and several antenna elements, which occupy the larger footprint along with power hungry components that lead to high cost and volume. mMIMO $O\text{-}RU_{7-2x}$ can be realized with different high level radio architecture based on various functional splits and interface/interconnects between these functional blocks.

2.5.1 $O\text{-}RU_{7-2x}$ Massive MIMO High-Level Functional Block Diagram

The level of integration achieved on a SoC for different functional blocks can vary and that in turn can result in different $O\text{-}RU_{7-2x}$ architecture and the associated interconnects between the functional blocks.

These functional splits can be realized through three possible configurations of $O\text{-}RU_{7-2x}$ mMIMO Architecture options and they are discussed in detail in the sections below. All configurations have the same three major functional blocks like O-RAN Fronthaul Processing Unit, Digital Processing Unit, and RF Processing unit.

For any $O\text{-}RU_{7-2x}$ mMIMO Architecture option, the eCPRI frame that comes from $O\text{-}DU_{7-2x}$ is compressed/decompressed and processed by the O-RAN Fronthaul Processing Unit & Beamformer block. The Digital Processing Unit, which has a FPGA/ASIC based DFE and AFE modules, receives the packets/digital signals (depending on the architecture used) from O-RAN Fronthaul Processing Unit & Beamformer block and performs the functionalities such as compression/decompression (if applicable), Low-PHY (if applicable), DDC, DUC, CFR, DPD, ADC and DAC. All these three functional blocks can be part of a single SoC or can be split into multiple physical integrated circuits (IC) and connected with a digital or packet-based interface. The functional block labelled RF processing block consists of an optional frequency converter (mixer), Power Amplifier (PA)/ Low Noise Amplifier (LNA). The filter is the next functional block in the path and is needed for each of the N Tx/Rx paths to meet the emissions in the DL and provide the necessary selectivity for UL. The last functional block in the mMIMO RU system is the Antenna. Depending on the power level for each of the Tx chain in a mMIMO system, the filter could be integrated into the RF Section or could be integrated along with the antenna thus making it the AFU. A detailed description of the O-RAN Fronthaul Processing, Digital, and RF Processing blocks is discussed in Section 2.3.2.

In a mMIMO O-RU, Tx/Rx reciprocity calibration is one of the important steps for optimal performance. Calibration is required to align the Tx and Rx chains in phase and amplitude. In some implementations, reciprocity calibration requires implementing a dedicated feedback path. Multiple options are possible for the design of such a feedback path – the main three options are shown in Figure 2.5.1-1, Figure 2.5.1-2, and Figure 2.5.1-3. The main difference between each of the calibration options is the implementation of the feedback network and where the RF feedback and associated distribution circuitry reside. The calibration signal ultimately needs to be relayed back to the AFE, so that the phase and amplitude difference between each of the corresponding Tx-Rx chain and the phase and amplitude differences between the multiple Tx-

Rx chains can be estimated and calibrated accurately to ensure uplink and downlink channel reciprocity to enable beamforming.

In Calibration option #1, the RF feedback and the distribution circuitry resides in the antenna module and all the RF components including the antenna are part of the Tx/Rx reciprocity calibration loop. In Calibration option #2, the RF feedback and the distribution circuitry resides in the filter module and all the RF components till the filter module are part of the Tx/Rx reciprocity calibration. In Calibration option #3, the RF feedback and the distribution circuitry resides in the RF Section and all the RF components till the RF Section's output in the Tx path and input in the Rx path are part of the Tx/Rx reciprocity calibration. These calibration options (CAL option #1, #2 and #3) can apply to all three mMIMO O-RU architectures in Figure 2.5.1-1, Figure 2.5.1-2, and Figure 2.5.1-3. It is also worth mentioning that other calibration options are possible where calibration feedback is not needed.

2.5.1.1 O-RU$_{7-2x}$ Massive MIMO High Level Architecture Option #1

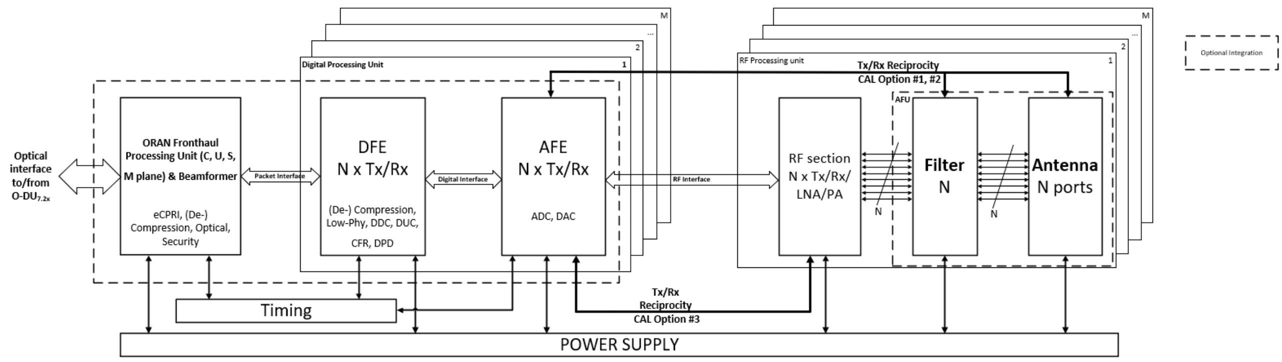


Figure 2.5.1-1 [NxM]T[NxM]R mMIMO O-RU$_{7-2x}$ High Level System Architecture #1

In this section, the diagram of NxM mMIMO O-RU$_{7-2x}$ High Level System Architecture option #1 is provided. This is a difference factor from the other architectures. In this example the Low-PHY functionalities are implemented in DFE block and ADC and DAC converters are in the AFE block, both of them inside Digital Processing Unit.

2.5.1.2 O-RU$_{7-2x}$ Massive MIMO High Level Architecture Option #2

In the mMIMO O-RU$_{7-2x}$ High Level System Architecture #2, the operations that are performed in DFE of mMIMO O-RU High Level System Architecture #1 are distributed between the O-RAN Fronthaul Processing Unit & Beamformer and AFE as shown in Figure 2.5.1-2. In particular, the Low-PHY functions of DFE are performed in the ASIC/FPGA for O-RAN Fronthaul Processing Unit & Beamformer and DDC, DUC, CFR, and DPD blocks of DFE are part of AFE in the RFIC as shown in Figure 2.5.1-2. The interconnection between the ORAN Fronthaul Processing Unit & Beamformer and Digital Processing Unit will be done through a digital interface since the Low-PHY is in the first block in this option.

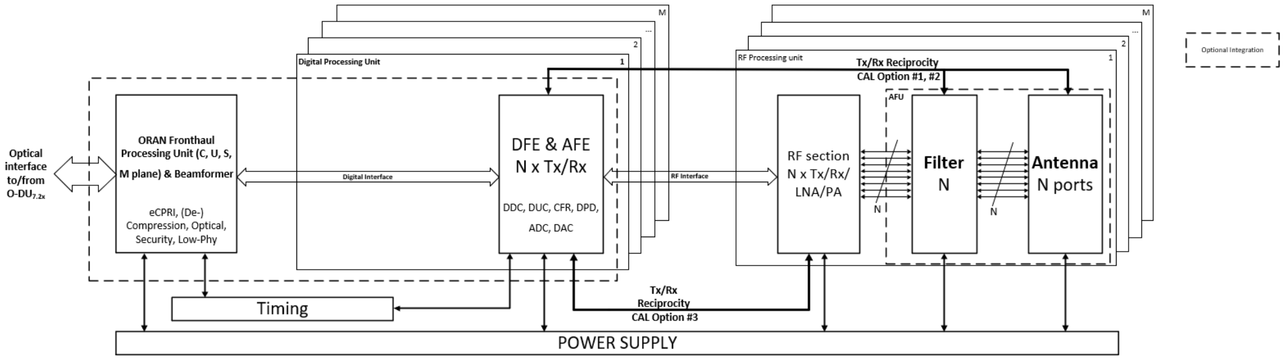


Figure 2.5.1-2 [NxM]T[NxM]R mMIMO O-RU$_{7\text{-}2x}$ High Level System Architecture #2

2.5.1.3 O-RU$_{7\text{-}2x}$ Massive MIMO High Level Architecture Option #3

In the mMIMO O-RU$_{7\text{-}2x}$ High Level System Architecture #3, DFE and AFE functions are combined on a single DFE & AFE functional module as shown in Figure 2.5.1-3. In this architecture, (De-)Compression, Low-PHY, DDC, DUC, CFR, and DPD are part of the same FPGA/ASIC. There is a packet-based interface/interconnect between the O-RAN Fronthaul Processing Unit & Beamformer and the DFE & AFE functional module. These modules can be on a single SoC or can be on different FPGA/ASIC.

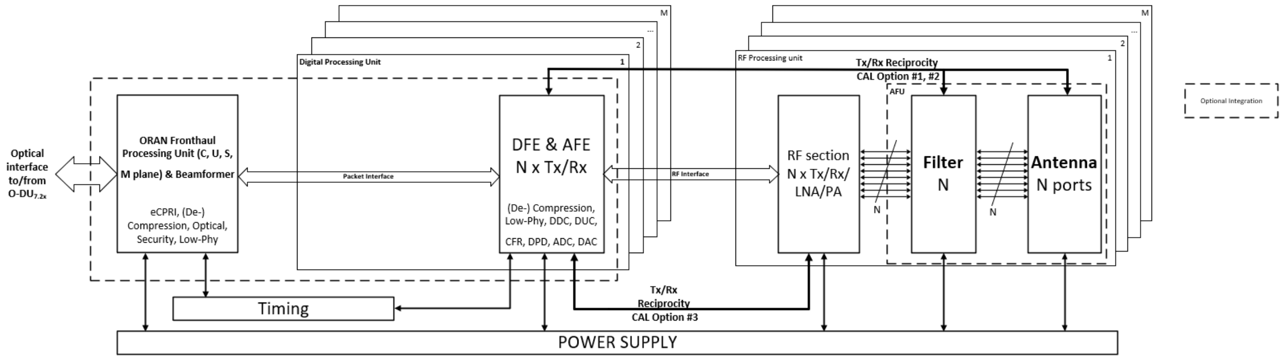


Figure 2.5.1-3 [NxM]T[NxM]R mMIMO O-RU$_{7\text{-}2x}$ High Level System Architecture #3

2.5.2 O-RU$_{7\text{-}2x}$ Massive MIMO Hardware Reference Design Architecture

The mMIMO O-RU$_{7\text{-}2x}$ hardware reference design architecture examples of 32T32R and 64T64R are shown in Figure 2.5.2-1 and Figure 2.5.2-2, respectively.

The 32T32R and 64T64R hardware architectures shown in Figure 2.5.2-1 and Figure 2.5.2-2, respectively, have four major functional blocks such as ORAN fronthaul processing, digital processing, RF processing (RF transceiver and RF front end units), and antenna units and Section 2.5.1 describes each of these blocks. The ORAN fronthaul processing unit is connected with the digital processing unit through a packet interface. the Digital processing unit is connected to the RF processing unit through a digital interface with an interface such as JESD204. In the RF processing block, the RF Transceiver communicates with the RF front end using an RF interface. The RF transceiver unit has four RFICs that cater for 32T32R i.e., thirty-two transceiver chains, where each RFIC contains eight transceiver chains as shown in Figure 2.5.2-1. Similarly, the RF transceiver unit of 64T64R shown in Figure 2.5.2-2 has eight RFICs. RF front end units are accordingly connected to the transceiver chains.

RF front end units connected with the antenna array with the 96 antenna elements, where each element is dual polarized (Vertical and Horizontal) thereby total number of antenna elements are 192. The antenna elements can be grouped in different ways. As an example, these elements can be arranged in two rows vertically and eight columns horizontally, where each row has 6 element sub arrays for 32T32R as shown in Figure 2.5.2-1. Similarly, for 64T64R, 96 antenna elements can be arranged in four rows vertically and eight columns horizontally, where each row has 3 element sub arrays as shown in Figure 2.5.2-2. The transceiver output can be either equally or not (following a specific criterion) fed to the all the subarray elements at the same time with the help of a power divider. Moreover, the antenna calibration process is described in detail in Section 2.5.1.

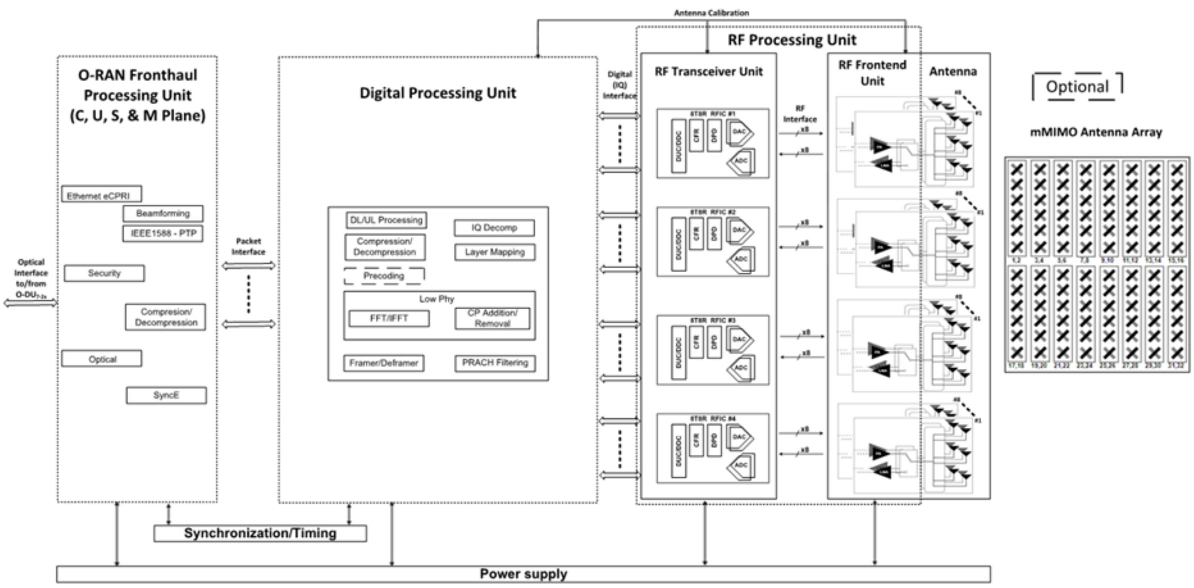


Figure 2.5.2-1 mMIMO O-RU$_{7\text{-}2x}$ hardware reference design architecture example of 32T32R

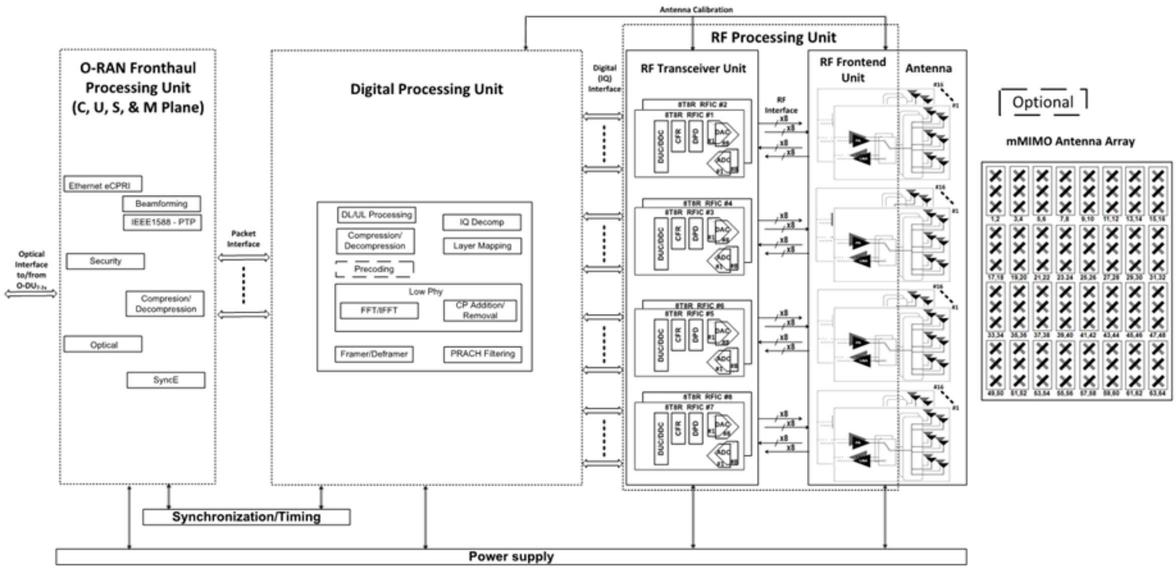


Figure 2.5.2-2 mMIMO O-RU$_{7-2x}$ hardware reference design architecture example of 64T64R

2.5.2.1 O-RAN Fronthaul Processing Unit

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2.5.2.2 Digital Processing Unit

A Digital processing unit (DPU) architecture example of massive MIMO O-RU$_{7-2x}$ is shown in Figure 2.5.2-3 and a detailed description of the same is discussed in Sections 2.5.1 and 2.5.2. The DPU processes the packets from O-RAN fronthaul processing unit through the packet interface and feeds the data/control mapped RF output signals to the RF frontend unit through the RF interface to amplify the RF signals according to the output power requirement of the signal chain. These are subsequently fed to the antenna thereby the signals can be received by the user equipment, e.g., mobile subscriber terminal, home CPE, Wi-Fi device etc...

The massive MIMO baseband processing requirements are largely based on the antenna data rate, Instantaneous/occupied bandwidth, number of transceiver chains, number of layers and component carriers and so on. Hence, the massive MIMO digital processing unit need to have one or more i.e., N number of digital front-end units to cater to the necessary baseband signal processing as shown in Figure 2.5.2-6 where an example implementation of the Digital Processing Unit is exposed; other implementations are not precluded.

The digital processing unit consists of the processing system and FPGA/ASIC blocks, where the later one handles the functions like compression/decompression, layer mapping, beamforming matrix generation/decoding, FFT/IFFT, CP removal/insertion, channel filtering, and many more. Apart from these, the digital processing unit performs PRACH, and SRS associated functions. Some of the digital processing units may also handle the RF transceiver functions like channel DUC/DDC, CFR, DPD, and DAC/ADC along with associated digital serial interfaces.

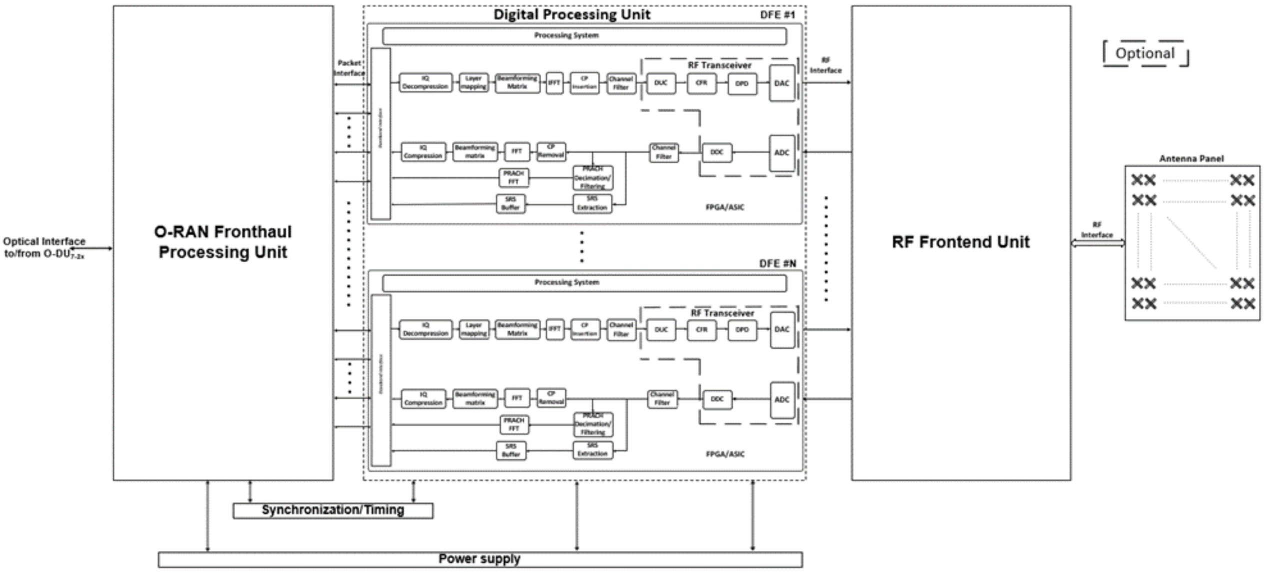


Figure 2.5.2-3 mMIMO O-RU$_{7\text{-}2x}$ Digital processing unit architecture example

2.5.2.3 RF Processing Unit

<<< This is left blank for the future contributions>>>

2.5.2.3.1 RF Transceiver Unit

<<< This is left blank for the future contributions>>>

2.5.2.3.2 RF Front End Unit

In general, the transmitter chain of mMIMO RF front end unit consists of balun, variable gain amplifier or step attenuator, amplifier stage (power amplifier, driver amplifier (optional), and pre-driver amplifier (optional)), circulator, and the filter (can be cavity or ceramic filters) as shown in Figure 2.5.2-4 and Figure 2.5.2-5 for 32T32R and 64T64R configurations, respectively. Similarly, the receiver chain consists of the low noise amplifier stage and band pass filter. The low noise amplifier stage consists of one or more low noise amplifiers either in normal or bypass mode depending on the received power level and the corresponding amplification requirement.

The Digital to analog converter (DAC) output (Downlink RF signal) of the RF transceiver unit is fed to the respective transmitter chain for the amplification and subsequently fed to the antenna. Similarly, the receiver chain (Uplink signal) output from the antenna is fed to the respective analog to digital convertor input of the RF transceiver unit after amplification by the low noise amplifier stage and filtering.

The number of RF transceiver units is decided based on the number of DAC outputs/ADC inputs that can be handled by the single unit. As an example, in Figure 2.5.2-4, for 32T32R configuration, four RF transceiver units are present, where each has eight ADC/DAC ports. Subsequently, thirty-two RF front end line-ups are connected to the respective DAC output of RF transceiver unit in the case of the transmitter chain and ADC input of RF transceiver in the case of receiver chain as shown in Figure 2.5.2-4.

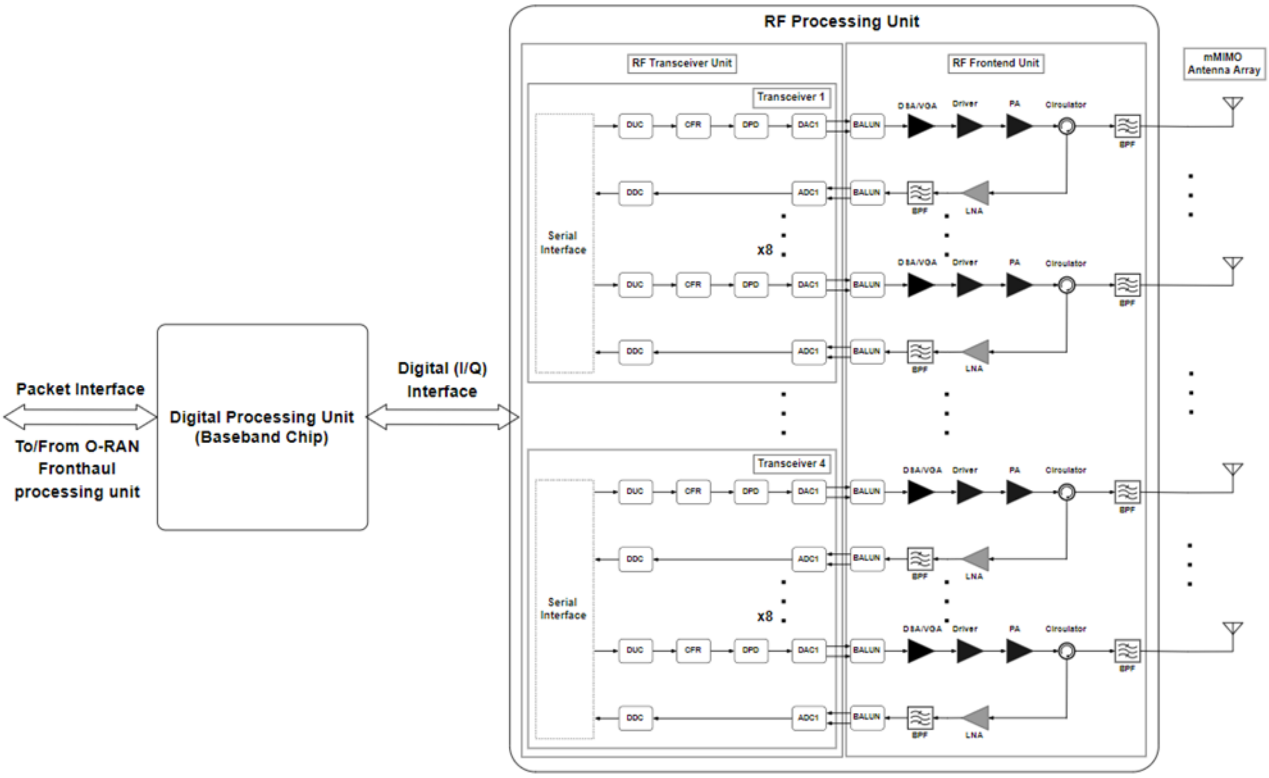


Figure 2.5.2-4 mMIMO O-RU$_{7\text{-}2x}$ RF front end architecture example of 32T32R

Like the 32T32R, as an example, in Figure 2.5.2-5, for 64T64R configuration, eight RF transceiver units are present, where each has eight ADC/DAC ports. Subsequently, sixty-four RF front end line-ups are connected to the respective DAC output of RF transceiver unit in the case of the transmitter chain and ADC input of RF transceiver in the case of receiver chain as shown in Figure 2.5.2-5.

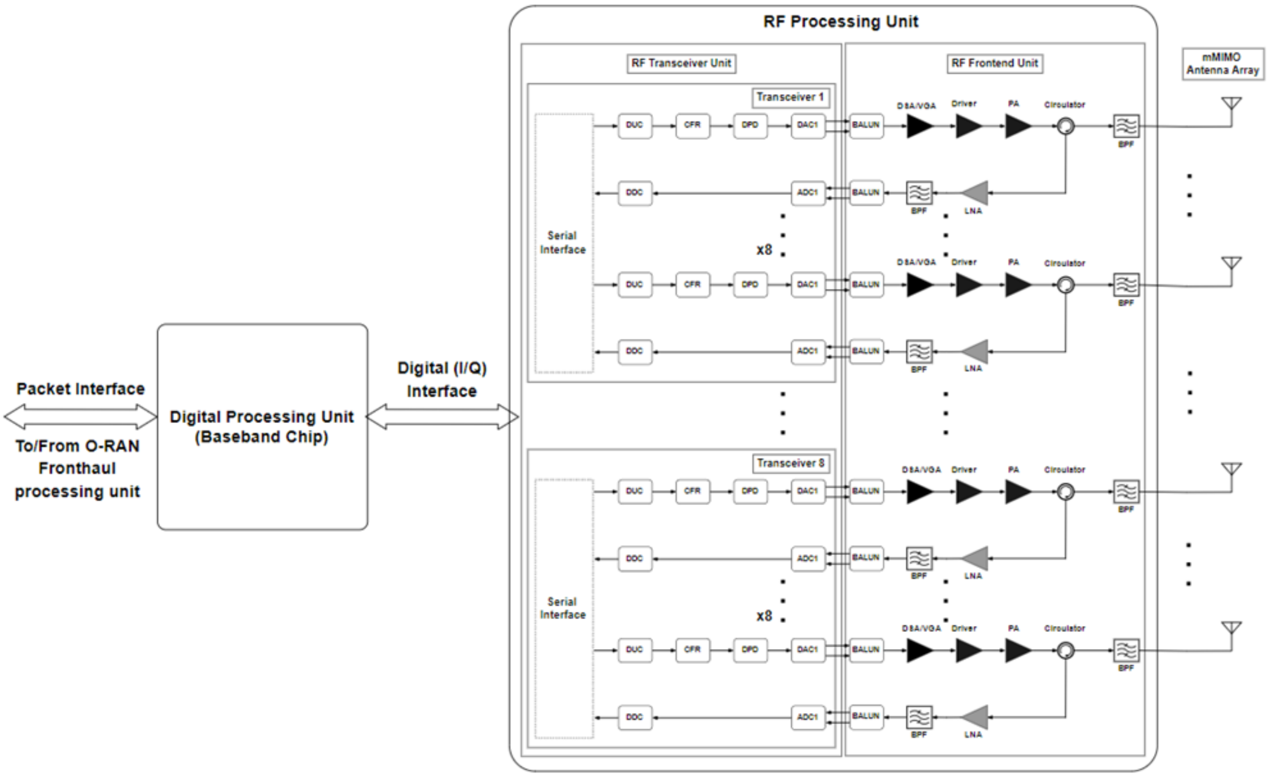


Figure 2.5.2-5 mMIMO O-RU$_{7\text{-}2x}$ RF front end architecture example of 64T64R

2.5.2.4 mMIMO Antenna/Phased array reference design

A mMIMO antenna array reference architecture examples for 32T32R and 64T64R are shown in Figure 2.5.2-6 and Figure 2.5.2-7 respectively. In Figure 2.5.2-6, two transceiver chains are connected to the one sub array with six dual polarized elements. i.e., one transceiver chain is connected to one polarization (vertical) sub array and another one with another polarization (horizontal) sub array. Similarly, each of the transceiver chains are connected to one of the two polarizations of sub arrays. The overall antenna element arrangement of an array for 32T32R is shown in Figure 2.5.2-6

In Figure 2.5.2-6 mMIMO O-RU7-2x antenna array architecture example of 32T32R the parameters like transmitted power per chain, antenna element gain, total antenna array gain, and finally the EIRP are shown/mentioned at the respective part of the architecture for 32T32R configuration.

The details of parameters are listed below along with the respective equations.

Total output power of the mMIMO Radio Unit in dBm, P$_T$ = $10\log_{10}\ (P_T, W)$ + 30, dBm,

Transmit power per chain, P$_c$ = $\frac{\boldsymbol{Total\ output\ power\ of\ the\ mMIMO\ Radio\ Unit\ (P_T),\ \ (W)}}{\boldsymbol{No\ of\ transceiver\ chains}}$

Transmit power per chain in dBm, P$_c$ = $10\log_{10}\ (P_c, W)$ + 30, dBm,

Antenna element gain = G_E, dBi,

Total antenna array gain = G_T, dBi,

Total antenna array absolute gain, G_T, abs = $\boldsymbol{G_E}, abs \; \times total \; no \; of \; elements,$

Total antenna gain in dBi, G_T = $10 \log_{10} \; (\boldsymbol{G_T}, abs)$, dBi,

EIRP = P_T, dBm+ G_T, dBi.

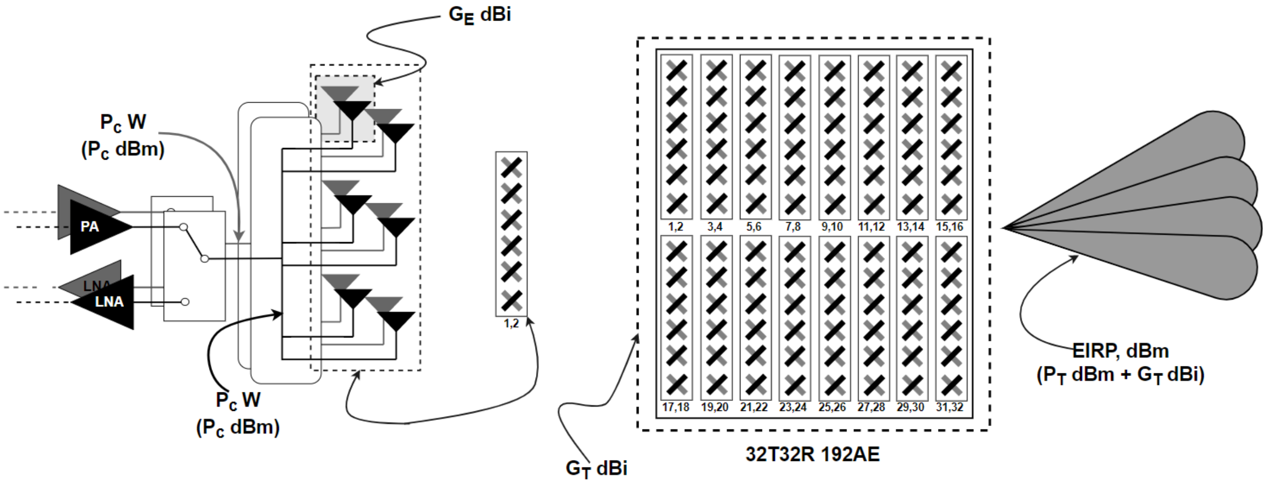


Figure 2.5.2-6 mMIMO O-RU$_{7\text{-}2x}$ antenna array architecture example of 32T32R

An example calculation for the 32T32R mMIMO in the entire RF chain till the antenna, i.e., EIRP is detailed out below.

Example calculation for 32T32R mMIMO configuration:

Specifications:

No of transceiver chains = 32 i.e., 32T32R,

Total output power, P_T = 320 W,

EIRP = 79 dBm,

Total number of antenna elements = 96 (*dual linear polarized, hence total number of elements would be 192*),

Number of elements in a subarray = 6.

Calculation:

Total output power P_T = 320 W,

Transmit power per chain, P_c = $\frac{320}{32}$ = 10 W,

Total output power of the mMIMO Radio Unit in dBm, P_T = $10 \log_{10} \; (320, W)$ + 30 = 55.05 dBm,

Transmit power per chain in dBm, P_c = $10\log_{10}(10\,W)$ + 30 dBm = 40 dBm,

Antenna element gain, G_E = 4 dBi,

Final antenna array absolute gain, G_T, abs = 2.51 x 96 = 240.96,

Antenna subarray gain in dBi, G_T = 23.82 dBi,

EIRP = 55.05 + 23.82 = 78.87 dBm,

Similarly, the parameters like transmitted power per chain, antenna element gain, sub array antenna gain, total antenna array gain, and finally the EIRP are shown/mentioned at the respective part of the architecture in Figure 2.5.2-7 mMIMO O-RU7-2x antenna array architecture example of 64T64R.

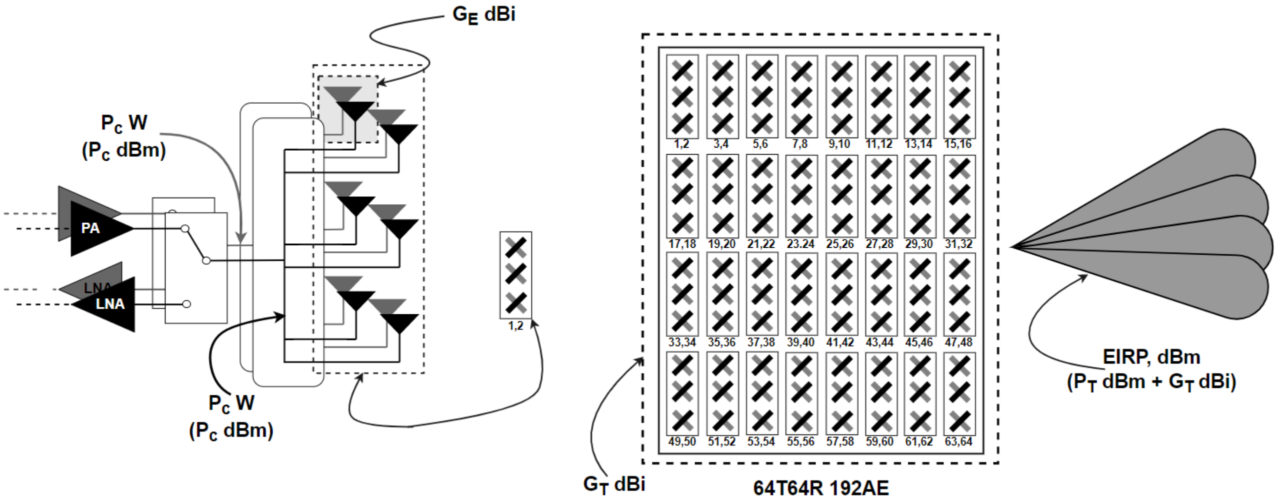


Figure 2.5.2-7 mMIMO O-RU$_{7\text{-}2x}$ antenna array architecture example of 64T64R

Also, the example calculation for the entire RF chain till antenna for 64T64R is presented below

Example calculation 64T64R mMIMO configuration:

Specifications:

No of transceiver chains = 64 i.e., 64T64R,

Total output power, P_T = 320 W,

EIRP = 79 dBm,

Total number of antenna elements = 96 (*dual linear polarized, hence total number of elements would be 192*),

Number of elements in a subarray = 3.

Calculation:

Total output power, P_T = 320 W,

Transmit power per chain, P_c = $\frac{320}{64}$ = 5 W,

Total output power of the mMIMO Radio Unit in dBm, P_T = $10\log_{10}\ (320, W)$ + 30 = 55.05 dBm,

Transmit power per chain in dBm, P_c = $10\log_{10}\ (5\ W)$ + 30 dBm = 37 dBm,

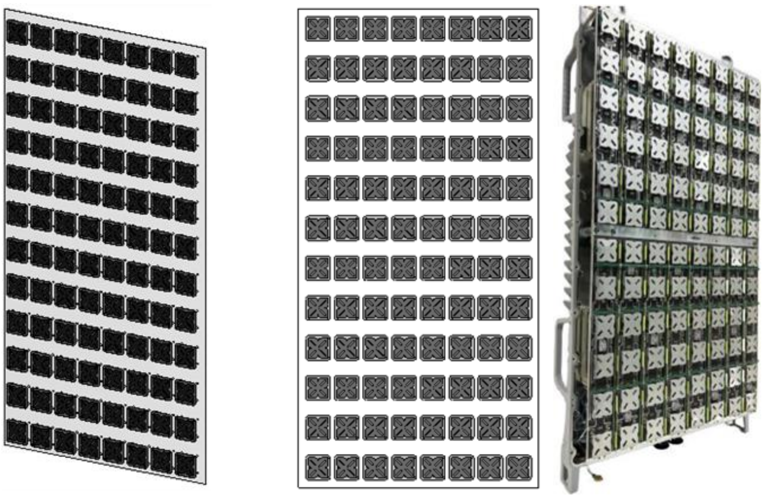
Antenna element gain, G_E = 4 dBi,

Final antenna array absolute gain, G_T, abs = 2.51 x 96 = 240.96,

Antenna subarray gain in dBi, G_T = 23.82 dBi,

EIRP = 55.05 + 23.82 = 78.87 dBm.

The simulation model of mMIMO antenna array with 192 reflector backed elements and the 32T32R mMIMO O-RU$_{7\text{-}2x}$ antenna array prototype is shown in Figure 2.5.2-8 mMIMO antenna array view (a) simulation model of 192 reflector backed antenna element array only and (b) 32T32R mMIMO O-RU7-2x Antenna array prototype [14], respectively.



(a) (b)

Figure 2.5.2-8 mMIMO antenna array view (a) simulation model of 192 reflector backed antenna element array only and (b) 32T32R mMIMO O-RU$_{7\text{-}2x}$ Antenna array prototype [14]

2.5.3 Mechanical

Various mechanical designs are possible for the mMIMO O-RU$_{7\text{-}2x}$. As an example, the isometric view of mMIMO O-RU$_{7\text{-}2x}$ is shown in Figure 2.5.3-1 Mechanical enclosure, where the heat sink fins and Radome are attached/placed at the backside and frontside of the radio unit, respectively. The heatsink fins are used to dissipate the heat that is generated inside the radio unit to the backward direction. The Radome is a RF transparent cover, and it acts as a shield and protect the antenna and radio unit from the environment while allowing the radiated waves from an antenna to the free space environment.

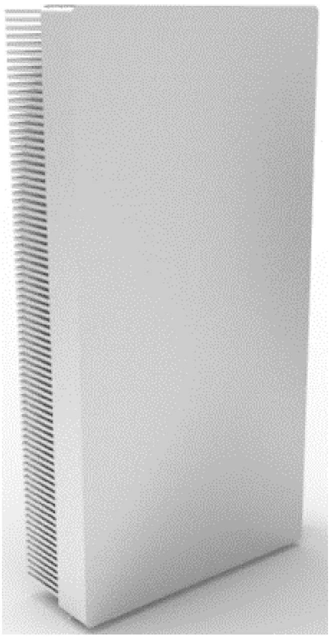
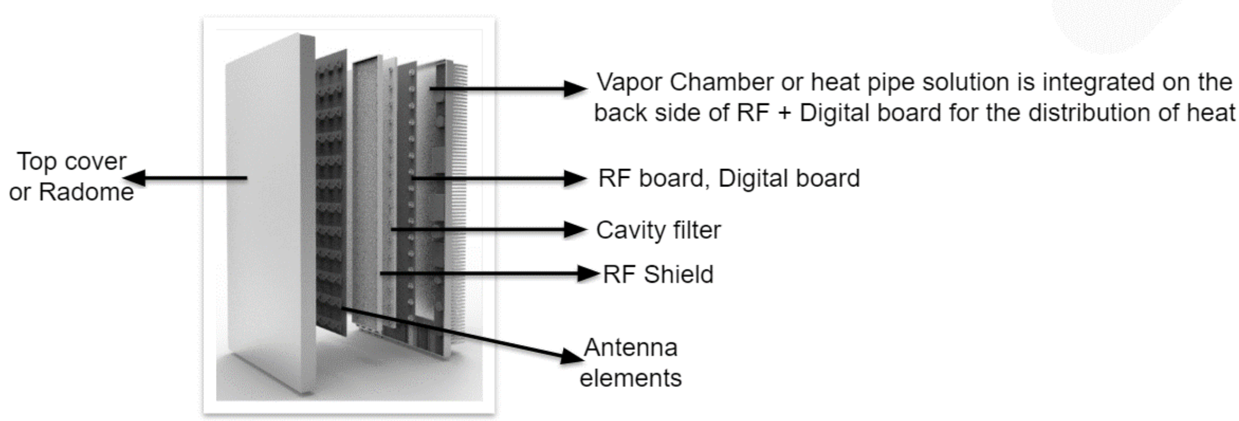


Figure 2.5.3-1 Mechanical enclosure

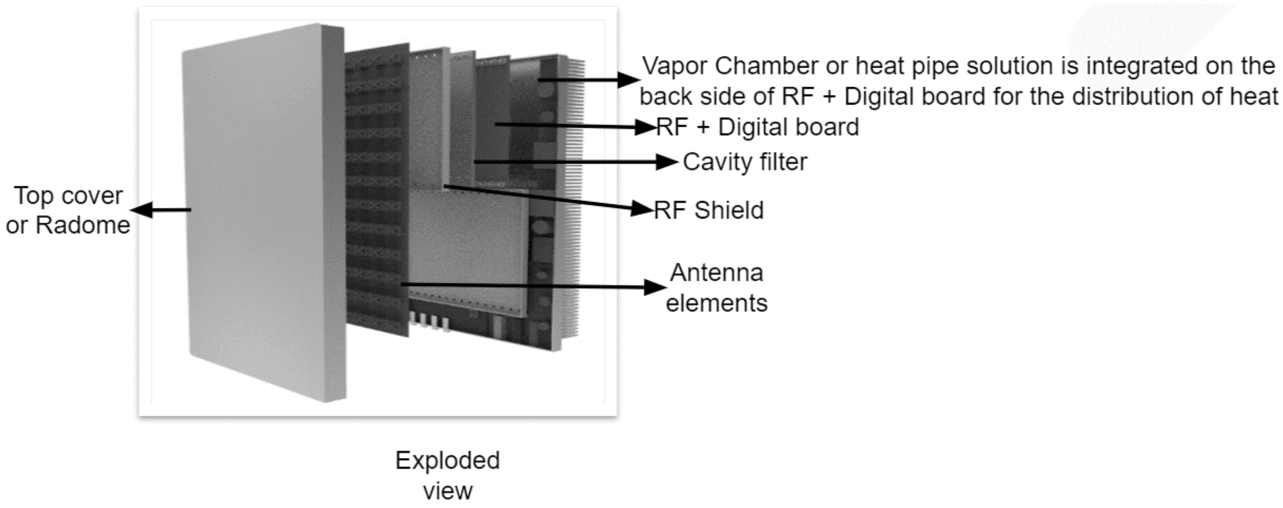
Figure 2.5.3-2 Exploded views of mechanical architecture/stacking examples (a) design #1 and (b) design #2 (a) and (b) show the exploded view of two mechanical design examples along with optional stacking. These figures illustrate the stacking of heat sink including vapor chamber or heat pipe for the heat dissipation, RF board, digital board, cavity filter, RF shield, antenna elements, and finally the top cover (Radome). These designs basically show the single stack up approach where all the RF chains, digital, and power section are placed in same heatsink surface area to make the design thermally stable one with few respective challenges. Different sections can be divided into multiple boards to reduce the complexity of routing as well as testing. Vapor chamber or heat pipe solution will help to mitigate this problem. If required, cut-outs can be given in the shield to provide heat sink area to major heat dissipating components.

Figure 2.5.3-2 Exploded views of mechanical architecture/stacking examples (a) design #1 and (b) design #2 shows the mechanical stacking/design where the components/peripherals placed vertically on the RF and digital boards, which results in large height along with moderate width. Accordingly, the cavity filter is designed in accordance with the respective connector position and RF shield also placed vertically to act as an isolator between the radiations coming from the RF board and antenna, in other words, it blocks the signals from RF board to antenna and vice versa. Antenna elements are placed as shown in Figure 2.5.3-2 Exploded views of mechanical architecture/stacking examples (a) design #1 and (b) design #2(a) and (b).

Figure 2.5.3-2 Exploded views of mechanical architecture/stacking examples (a) design #1 and (b) design #2 the stacking option where components/peripherals of RF and digital boards are placed horizontally, and this results in the overall radio unit to have a broader width and moderate height. Here the RF shield and cavity filters are placed in accordance with the layout of RF and digital boards.



(a)



(b)

Figure 2.5.3-2 Exploded views of mechanical architecture/stacking examples (a) design #1 and (b) design #2

2.6 FHGW Hardware Reference Design

The O-RAN Hardware Reference Design Specification for Fronthaul Gateway v2.0 (if required by design) can be found here: https://orandownloadsweb.azurewebsites.net/specifications

The fronthaul gateway can be used to terminate O-RAN fronthaul user, control, synchronization, and management planes for option 7-2x and to translate the signals to and from several CPRI interfaces (option 8). It can further multiplex or demultiplex option 7-2x fronthaul signals to or from several O-RUs. This version of the specification only supports option 7-2x to option 8 conversion and vice versa in addition to the transport of other traffic streams.

Annex 1: Parts Reference List

According to WG7 scope and charter, component selection, for example implementation of white box hardware is allowed for WG7 but would not be mandatory in any Specification.

1.1 Supermicro O-DU:

No.	Supermicro Part Number	Description
1	SYS-112D-42C-FRDN8P	X14 1U 400mm Rackmount Server, with dual 600W DC Power Supply
2	(integrated SOC in above part)	Intel Xeon 6 Granite Rapids D 42core CPU
3	MEM-DR564L-SL02-ER56	64GB DDR5-5600 2Rx4 LP (16Gb) ECC RDIMM
4	HDS-MMN-MTFDKBA480TFR1BC	Micron 7450 PRO 480GB NVMe PCIe4.0 M.2
5	AOM-TPM-9670V-S-O	Trusted Platform Module 2.0

1.2 ADI Based O-RU 4T4R:

No.	Item Name	Description
1	ADP196ACBZ-R7	5V, 3A Logic Controlled High-Side Power Switch
2	LTC7150SEY#PBF	20V, 20A Synchronous Ste-Down Regulator
3	ADRV9029	Integrated, Quad RF Transceiver with Observation Path and DPD/CFR
4	ADP223ACPZ-R7	Dual, 300 mA Adjustable Output, Low Noise, High PSRR Voltage Regulator
5	LTC4217IDHC#PBF	2A Integrated Hot Swap Controller
6	ADCLK846BCPZ-REEL7	1.8V, 6 LVDS/12 CMOS Output Low Power Clock Fanout Buffer
7	AD9545BCPZ	IEEE 1588 V2 and 1 pps Synchronizer and Adaptive Clock Translator
8	ADRF5545A	Dual-Channel 2.4 to 4.2 GHz Receiver Front End
9	LTC3315AIV#TRMPBF	Dual 5V, 2A Synchronous Step-Down DC/DCs
10	LTC3634EUFD#PBF	15V Dual 3A Monolithic Step-Down Regulator for DDR Power
11	LTC2928IUHF#BPF	Multichannel Power Supply Sequencer and Supervisor
12	ADP5054ACPZ-R7	Quad Buck Regulator Integrated Power Solution
13	AD9576BPZ	Dual PLL, Asynchronous Clock Generator
14	LT1389ACS8-1.25#PBF	Nanopower Precision Shunt Voltage Reference
15	Intel® Arria® 10 SoC	Intel® Arria® 10 SoC 1AS048F35
16	MAX1619MEE	Temp Sensor with Dual-Alarm Outputs and PMBus

1.3 ADI Based O-RU 8T8R:

No.	Item Name	Description
1	AD9576BPZ	Dual PLL, Asynchronous Clock Generator
2	AD9545BCPZ	IEEE 1588 V2 and 1 pps Synchronizer and Adaptive Clock Translator
5	MAX6581TG9A+	±1°C Accurate 8-Channel Temperature Sensor
6	DS1339C-33#	I²C Serial Real-Time Clock
7	ADIN1300CCPZ	Robust, Industrial, Low Latency and Low Power 10 Mbps, 100 Mbps, and 1 Gbps Ethernet PHY
8	ADRV904x	8T8R 400MHz RF Transceiver, Integrated Observation Paths with DFE
9	Intel® AGFB014R24B212V	Intel® Agilex® AGF014
10	LTC2980	PMBus Power System Manager
12	LTC4251BIS6-1#TRMPBF	Negative Voltage Hot Swap Controllers in SOT-23
13	LTC3888-1	Dual Loop 8-Phase Step-Down DC/DC Controller with Digital Power System Management
14	LTC8625SP-1	18V/8A Step-Down Silent Switcher 3 with Ultralow Noise
15	LT8627SPJV#PBF	18V/16A Step-Down Silent Switcher 3 with Ultralow Noise
16	LTC3315AJV#WTRPBF	Dual 5V, 2A Synchronous Step-Down DC/DCs in Tiny LQFN and WLCSP
17	LTC3636EUFD#TRPBF	Dual Channel 6A, 20V Monolithic Synchronous Step-Down Regulator
18	LTC3618EUF#PBF	Dual 4MHz, ±3A Synchronous Buck Converter for DDR Termination
19	ADCLK846BCPZ-REEL7	1.8V, 6 LVDS/12 CMOS Output Low Power Clock Fanout Buffer

History

Date	Revision	Description
2022/02/02	v01.00	Published as final version 01.00
2022/11/09	v02.00	Published as revision 02.00 (adds a new section for O-RU internal interfaces with updated diagrams and massive MIMO reference design)
2023/02/16	v03.00	Published as revision 03.00 (updated section 2.2 O-DU spec and Annex 1.1)
2025/06/04	v04.00	Published as revision 04.00 (updated section 2.2 O-DU spec and Annex 1.1)

